



Machine learning in occupational accident analysis: A review using science mapping approach with citation network analysis



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ABSTRACT

The present study reviews the publications that examine the application of machine learning (ML) approaches in occupational accident analysis. The review process includes four phases of analysis, namely bibliometric search, descriptive analysis, scientometric analysis, and citation network analysis (CNA). In the bibliometric search, a total of 232 articles are systematically screened out from 1995 to 2019 (up to May). Then, descriptive analysis and scientometric analysis are carried out to find the influences of journals, authors, authors' keywords, articles/documents, and countries/regions in developing the domain. Thereafter, CNA is carried out to classify the publications according to the research themes and methods used. From this extensive review, several key findings are obtained in the application of ML approaches in occupational accident analysis. USA, China, and Taiwan are the leading countries/regions in publishing articles. The four major research domains are (i) prediction of incident outcomes, (ii) extraction of rule based patterns, (iii) prediction of injury risk, and (iv) prediction of injury severity. Then, a taxonomy of the ML algorithms used is developed. Finally, research gaps and safety issues are highlighted and the scope for future is discussed.

1. Introduction

According to the International Labour Organization (ILO) estimation, globally about 2.3 million workers succumb to death annually due to occupational accidents and diseases which include approximately 360 thousands fatal accidents (Sánchez et al., 2011). Overall, nearly 337 million occupational accidents are reported per year. From the ILO report, it is revealed that approximately 4% of the annual gross domestic product (GDP), which is equivalent to the US \$ 1.25 trillion, is drained off due to occupational accidents (ILO, 2008). Although the Government of different countries put much more efforts to curve down the occurrences of accidents, still they are happening frequently. Since these occurrences critically impact operations of organizations and generate economic burden to society, research on the occupational accident is much needed of late.

To research on occupational accidents, there exist several approaches, for example, conventional statistical analysis, survey-based qualitative analysis, data-driven machine learning (ML)-based analysis. Of them, ML-based analysis is rather a new approach, which attains more popularity over existing ones. With the emergence of a large volume of data, ML approach outperforms its traditional counterpart due to its some potential benefits, for example, (i) it is capable to deal

with large dimensional data, (ii) it is flexible in reproducing the data generation structure irrespective of complexity, and (iii) its predictive and interpretative potential through the extraction of relations/rules among the attributes in data (Martin et al., 2009; Matias et al., 2008). Due to these benefits, ML-based approaches have been used in diverse filed of applications, such as engineering, medical science, finance, and so forth (Witten et al., 2016). However, its use in occupational accident analysis has drawn relatively less attention of researchers so far. Therefore, it becomes essential to examine and classify the existing ML research on occupational accident domain and to extract the knowledge of their use in this domain. This ultimately puts forward the thrust of the future research trends of ML approaches in this domain.

1.1. Motivation and objectives

Based on the growing interest of researchers on ML approaches in occupational accident domain, there remains a gap. There exist a few review papers related to occupational accidents, for examples, Swuste et al. (2010) focused safety metaphors and theories in safety literature of the UK, US, and the Netherlands, McKenzie et al. (2010) explored the utility of text analysis in injury surveillance research through a systematic review, Khanzode et al. (2012) presented a comprehensive

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review of the concepts of occupational injury and accident causation and prevention, Samano-Rios et al. (2019) presented a review on occupational safety and health interventions to protect workers from hazards at work, Duarte et al. (2019) reviewed occupational accidents in the mining industry. Although these studies added values to the existing body of knowledge, the focus of those studies is limited in a particular topic, rather than considering the overall occupational accident research. Moreover, the application of ML approaches in this domain has not been reported as of now. Therefore, there is no such comprehensive review reported so far that can bring a broader picture of ML applications on this domain. To comprehend the shortcomings of the existing body of literature, the present study proposes a four-phase literature review on the application of ML approaches in occupational accident analysis. The four phases of analysis are bibliometric search, descriptive analysis, scientometric analysis, and citation network analysis. Therefore, in particular, the objectives of this review include: (i) applying bibliometric search to filter the relevant articles within a certain period; (ii) applying descriptive analysis to find the research trend in this domain in terms of journals, data collection method, machine learning methods, authors' contribution, contributions of institutes, countries/regions, sponsoring agencies, and different industry sectors; (iii) applying a scientometric analysis to find the influences of journals, authors, authors' keywords, articles/documents, and countries/regions in this domain; and (iv) applying the citation network analysis (CNA) to classify the overall research domain according to the research themes and methods used. Finally, the study explores several research issues and provides important directions for future research works in this domain.

The remainder of the paper is organized as follows: in Section 2, the four-phase methodology is explained briefly. In Section 3, the results of the four-phase analyses are presented and discussed. In addition, research growth is presented in this section. Research issues and the scopes for future research works are provided in Section 4. Finally, in Section 5, a conclusion with the limitations of the study is presented.

2. Research methodology

In this section, the research methodology of our review process is explained and depicted in Fig. 1. The main purpose of this review is to explore the wide range of applications of ML techniques in occupational accident research, to classify the overall research domain, and finally, to identify the research scopes in each domain. We limit our collection process of the articles of high scientific value using a structured four-phase process, namely (i) bibliometric search, (ii) descriptive analysis, (iii) scientometric analysis, and (iv) citation network analysis. The phases are briefly discussed below.

(i) Phase 1: Bibliometric search

- (a) **Time horizon:** The articles selected for this review process were published between 1995 and 2019 (up to May). The year 1995 is selected as the starting point to because the application of ML in occupational incidents is observed to be started from this year.
- (b) **Selection of database:** In this survey of the literature, papers were searched from Scopus database since it covered almost all journals and publications as compared to other digital sources (Aghaei Chadegani et al., 2013). In addition to this, other databases, namely Science Direct, IEEE Explore, and Springer-linked journals were also searched to identify any paper whether missed out or not in searching only from Scopus.
- (c) **Selection of journal:** In this study, the journals focusing on relevant domains, including engineering, safety, decision sciences, business management or computer science are considered since they publish articles in line with our focus of the survey. Some areas, including ergonomics and environmental hygiene are not included in this study as the objective of the

papers related to these areas do not match with that of our present study. This process has enabled us to finally screen out a total of 232 peer-reviewed journal papers for further analysis.

- (d) **Selection of articles:** During the searching of articles, the following title and keyword search functions are used: Title ("Occupational accidents") OR ("Safety Management") OR ("Safety") OR Keywords ("Incidents") AND Keywords ("Machine Learning") OR Keywords ("Data mining"). Initially, upon searching by title, a total of 1322 articles are found out. Thereafter, the keyword search function is used. Finally, the English-written articles published only in peer-reviewed journals are considered in this review process, which excludes a number of relevant papers published either as conference proceedings or as book chapters. This is done since they are published in a large quantity having comparatively less useful information in comparison with the articles published in journal papers (Butler and Visser, 2006). As a consequence, a set of 232 articles are extracted from the database. The articles, which are not directly based on ML application but are relevant to ML, are also considered in this study.

After data collection, a particular strategy was followed for data pre-processing. First, as a basic database, we considered Scopus. All the relevant papers from Scopus were initially retrieved in the form of.csv file format. We fixed the format for our final analysis. Following the information within this format, we prepared a similar.csv file from the data collected from the rest of the other databases. Finally, when all information were collected, all.csv files were merged, given that all of these files are identical, which means that all files have the same number of attributes. Using this merged dataset, further analyses, i.e., descriptive analyses, Scientometric analyses, and CNA were carried out.

(ii) Phase 2: Descriptive analysis

After the bibliometric search, a descriptive analysis is carried out to identify (i) the distribution of the number of articles by (i) the year of publications, (ii) the journals, (iii) the data collection methods, (iv) the distribution of methods, (v) the authors' contribution, (vi) the contribution of different institutes, (vi) the contribution of different countries/regions, (vii) the contribution of sponsoring agencies, and (vii) the contribution of different industries/sectors.

(iii) Phase 3: Scientometric analysis

After descriptive analysis, a scientometric study is carried out using VOSViewer, a software tool (van Eck and Waltman, 2009). The tool is capable of generating a distance-based visual representation of networks, where each distance between two nodes denotes the degree of their closeness. Using this approach, the influences/impacts of articles, journals, authors, and keywords in our research domain can be identified.

(iv) Phase 4: Citation network analysis

After Scientometric study, the citation network analysis (CNA) is done for the classification of the research domain. The advantages of CNA include that it is capable of classifying research domains and helping in obtaining research growth or changing paradigm (Colicchia and Strozzi, 2012; Ngai et al., 2008). In this study, the references and citations of the collected articles are first recorded, and then, the binary matrices are formed. Using these matrices, citation networks are developed in the form of clusters. In this study, modularity-based clustering approach is adopted (Newman and Girvan, 2004; Waltman et al., 2010).

Consider a network having n number of nodes. The objective is the generation of a mapping of these nodes. Let $link_{ij}$ indicates the number of links between nodes i and j , where $link_{ij} = link_{ji} \geq 0$. Let str_{ij} indicates the strength of link or association between the nodes i and j can be expressed by the following Eqs. (1) (Van Eck and

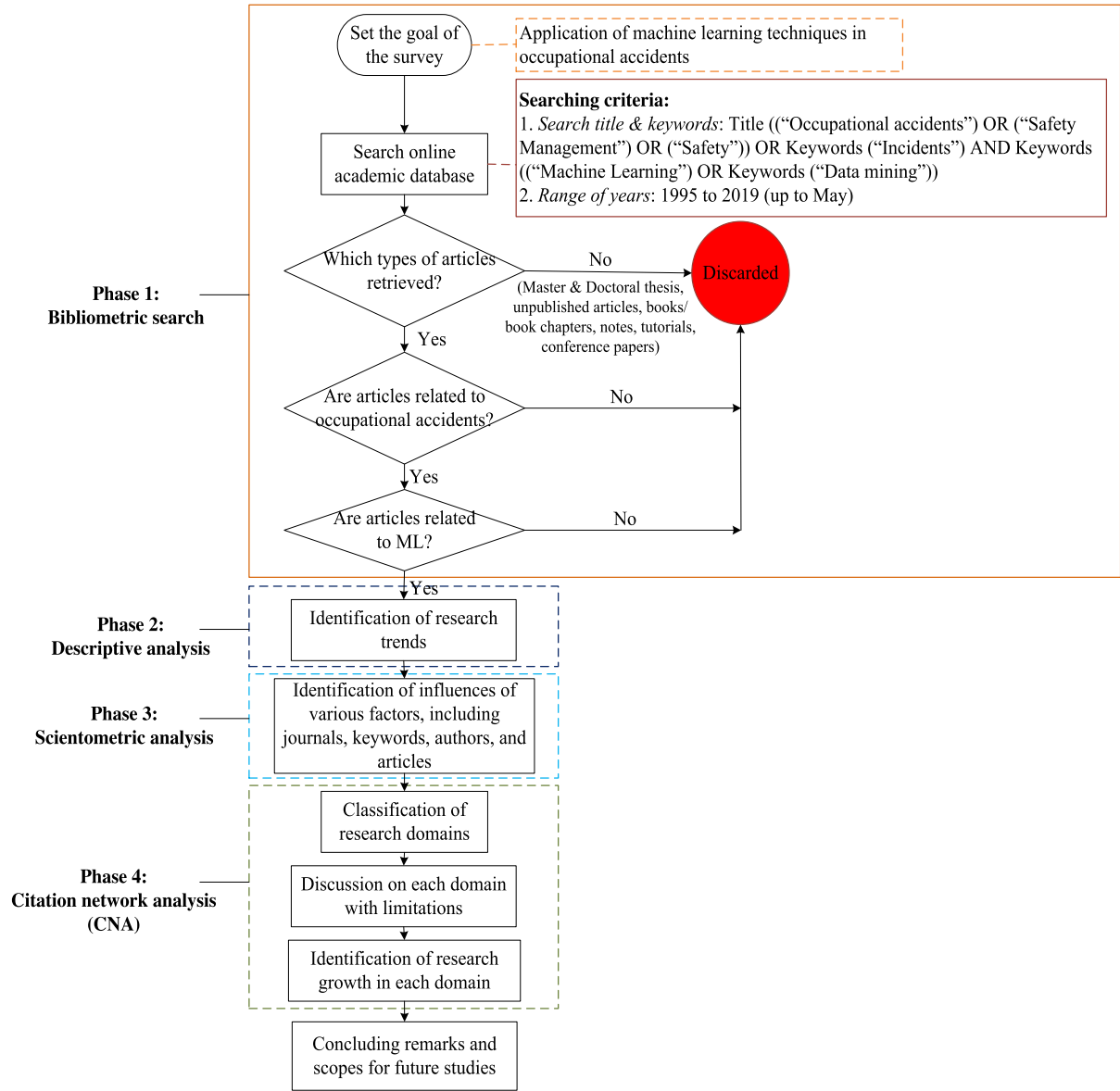


Fig. 1. Flowchart of the four-phase literature review process.

Waltman, 2007):

$$str_{ij} = \frac{2 \times total_{link} \times link_{ij}}{link_i \times link_j} \quad (1)$$

where $link_i$ represents the total number of links of node i and $total_{link}$ denotes the total number of links within the network. Here, $link_i$ and $total_{link}$ can be expressed by the following Eq. (2), and (3), respectively:

$$link_i = \sum_{j \neq i} link_{ij} \quad (2)$$

$$total_{link} = \frac{1}{2} \sum_i link_i \quad (3)$$

Now, we need to find a vector $vec_i \in R^{b'}$ for each node i . The vector represents the location of the node i in a b' -dimensional feature map. We require to find an integer (positive) value vec_i of for each of the nodes. Here, vec_i denotes the cluster in which the node i belongs to. The both clustering and mapping approaches are based

on minimizing the following expression given by Eq. (4):

$$V' \left(vec_1, vec_2, \dots, vec_{nd} \right) = \sum_{i < j} str_{ij} dist_{ij}^2 - \sum_{i < j} dist_{ij} \quad (4)$$

where $dist_{ij}$ indicates the distance between the nodes i and j . For mapping, the $dist_{ij}$ can be expressed by the following Eq. (5):

$$dist_{ij} = \|vec_i - vec_j\| = \sqrt{\sum_{k=1}^{b'} vec_{ik} - vec_{jk}}^2 \quad (5)$$

and for clustering, it is expressed by the following Eq. (6):

$$dist_{ij} = \begin{cases} 0, & \text{if } vec_i = vec_j \\ \frac{1}{\sigma}, & \text{if } vec_i \neq vec_j \end{cases} \quad (6)$$

Here, the parameter σ (where $\sigma > 0$) denotes a resolution parameter. It controls the number of clusters; if it increases, the number of clusters also increases. The Eq. (4) can be represented in the form of repulsive and attractive forces between the nodes. An attraction force and repulsive force is represented by the first and

second term of Eq. (4), respectively. The higher association strength indicates the stronger attraction force between nodes. For mapping, the above-mentioned can be considered as the mapping by VOSviewer (Van Eck and Waltman, 2007; Waltman et al., 2010), which is deeply related to the multi-dimensional scaling. For clustering, it can be shown that minimizing Eq. (4) is equal to maximizing the following expression given by Eq. (7):

$$\hat{v} \left(VEC_1, VEC_2, \dots, VEC_{nd} \right) = \frac{1}{2total_{link}} \sum_{i < j} \delta' \left(VEC_i, VEC_j \right) \tilde{w}_{ij} \left(link_{ij} - \delta' \frac{link_i \times link_j}{2total_{link}} \right) \quad (7)$$

where $\delta'(VEC_i, VEC_j)$ is equal to 1 if $VEC_i = VEC_j$ and zero, otherwise (Waltman et al., 2010). The weights $\tilde{w}_{ij}$ can be expressed by the following Eq. (8):

$$\tilde{w}_{ij} = \frac{2total_{link}}{link_i \times link_j} \quad (8)$$

If the parameters δ and the weights w_{ij} are equal to 1 in Eq. (7), the equation becomes a modularity function (Newman and Girvan, 2004). In scientometric and bibliometric studies, modularity-based clustering is used in some of the earlier studies (Schubert and Soos, 2010; Wallace et al., 2009; Zhang et al., 2010).

3. Results and discussion

In this section, results from descriptive analysis, scientometric analysis, and citation network analysis are discussed. Following this, the growth of research both in research domains and ML applications is discussed. Finally, some important observations from this research growth are summarized.

3.1. Descriptive analysis

In this section, the descriptive analyses are done to explore the distribution of the number of articles by year of publications, by journals, by data collection method, by the methods, by authors' contribution, by different institutes, by countries, by sponsoring agencies, and by industries or sectors or domains. This analysis provides an overall view of the current status or trend of ML applications in occupational accident research.

3.1.1. Distribution of the number of articles by year of publication

The number of articles published in the domain of ML-based occupational incident analysis from 1995 to 2019 (up to May) have been investigated. The outputs from this investigation are shown in Fig. 2. It is revealed from the figure that the application of ML is gradually gaining the attention of researchers, practitioners, academicians across

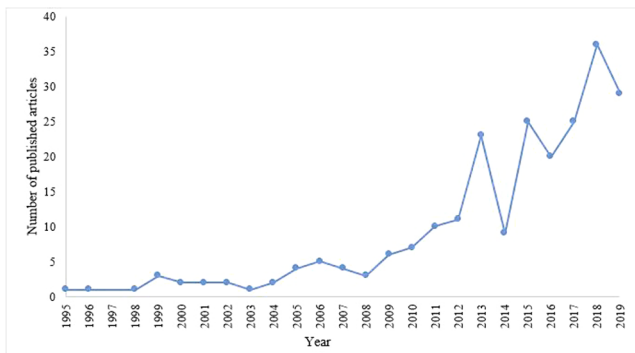


Fig. 2. All published articles from 1995 to 2019 (up to May).

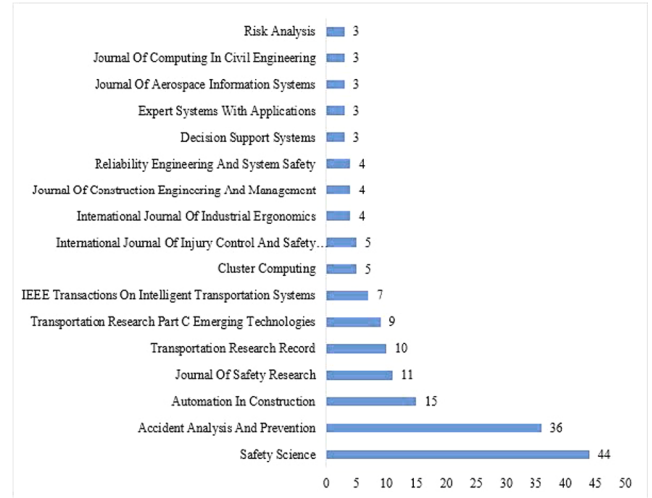


Fig. 3. No. of published articles in different journals (1995–2019) (max. 3 number of articles).

the world. The research traces back in the year 1995 with a very small number of publications and shows an increasing trend up to 2019 (May). However, in some years like 2014, 2016 reveals a drop in research contribution as compared to its previous year.

3.1.2. Distribution of the number of articles published by journals

The number of articles published in different journals from 1995 to 2019 (up to May) in this domain have been investigated. The outputs from this investigation are shown in Fig. 3. It is evident that 'Safety Science' has published the maximum number of articles during this period in the above-mentioned domain. It contributed 44 out of 232 articles, which is equal to 18.97% of the total articles collected. Following it, the journals 'Accident Analysis & Prevention', 'Automation in Construction', 'Journal of Safety Research' contributed 36, 15, and 11 articles, respectively. The journals, 'Risk Analysis', 'Reliability Engineering and System Safety', though aligned to the occupational accident research, have received little attention during this period.

3.1.3. Distribution of articles by data collection method

The distribution of data collection methods has been investigated. The outputs from this investigation shown in Fig. 4 indicate that 93.97% of 232 articles used Archival data, 2.16% articles used survey data (i.e., report published by different agencies), 1.72% articles used real-time data, and 0.96% articles used multi-source data. Survey data based on experiments counts as low as 0.86%. It is to be noted that archival data here represent the data collected prior to the analysis and are stored in an electronic database or files. They are collected once an event is completed. So, they cannot be changed and are sometimes

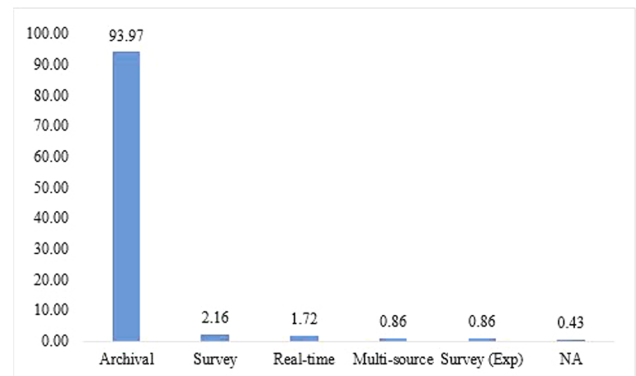


Fig. 4. Distribution of data collection methods.

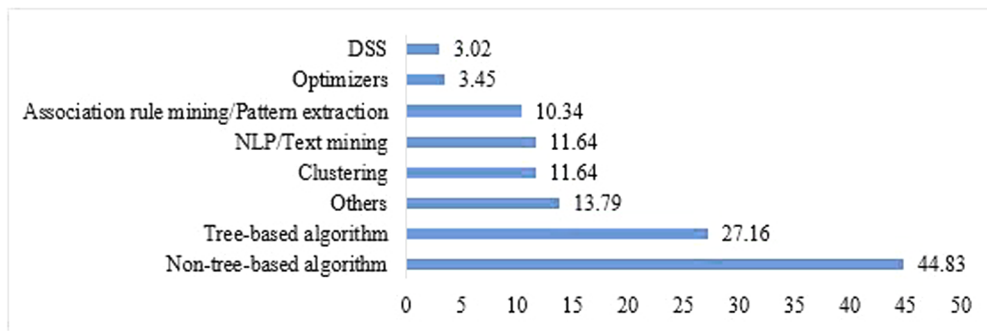


Fig. 5. Distribution of methods.

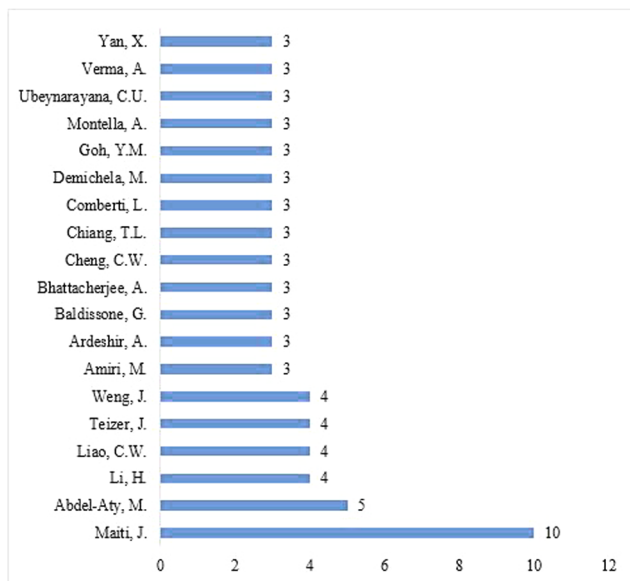


Fig. 6. No. of published articles by different authors (max. 3 number of publications).

called as fixed data. They are the secondary source of data. The survey data are the primary source of data. These data are usually collected from a sample of respondents. This type of data holds comprehensive information gathered from a targeted audience about a particular topic of interest to conduct a research. This type of data can be collected from multiple sources, such as online surveys, face-to-face surveys, telephonic surveys, etc.

3.1.4. Distribution of articles by the methods

The distribution of various methods has been also studied in this domain. The results from our investigations are displayed in Fig. 5. It is explored from this figure that 'Non-tree-based algorithms' are used mostly with 44.83% contribution in this domain. Here, 'Non-tree-based algorithms' imply the algorithms used for the classification task, for example, support vector machine (SVM), artificial neural network (ANN), extreme learning machine (ELM), Bayesian Network (BN), Naive Bayes (NB), and others. In contrast, 'Tree-based algorithms' are used with 27.16% contribution in this domain. Different algorithms in this category are deployed, for instances, C5.0, classification and regression tree (CART), random forest (RF). The other approaches, 'Clustering', which is an unsupervised learning approach, has been used 11.64% with 'Text mining/NLP' applications. The other important approach, namely 'Association rule mining' used for pattern extraction is applied with 10.34% contribution. More interestingly, it should be observed that optimization techniques also used in this domain with

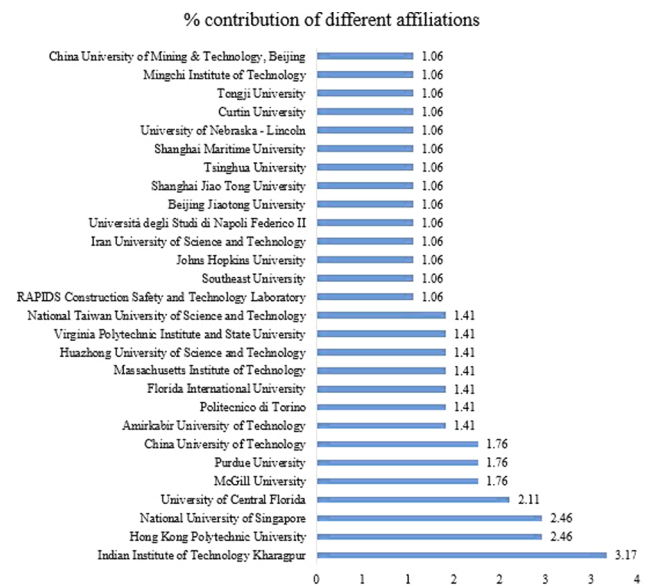


Fig. 7. No. of published articles in different institutes (max. 1.06% contribution).

3.45% contribution. Another important facet is the emergence of a support system, called DSS, which is used in this domain or related domain with 3.02% contribution.

3.1.5. Distribution of articles by authors' contribution

The contribution of the individual author or researcher has been analysed in this domain. The outputs from this investigation are shown in Fig. 6. The contribution of the researchers is shown as minimum as three number of publications. In this figure, it is evident that 'Maiti, J.' from IIT Kharagpur, India is ranked top in the list with ten number of publications. Following him, 'Abdel-Aty, M.' (with five number of publications) from the University of Central Florida, Orlando, United States, 'Li, Heng' (with four number of publications) from the Hong Kong Polytechnic University (HKPU) are ranked in the list.

3.1.6. Distribution of articles by different institutes

It is also investigated the contributions of different institutes in this domain. The results from our investigations are depicted in Fig. 7. In this figure, the contributions (up to 1.06% contribution of publications) of different institutes are shown. It is explored that 'IIT Kharagpur' in India is ranked top in the list with 3.17% contribution of publications. 'HKPU' in Hong Kong ties its position with the 'National University of Singapore' in Singapore for the second position in the list with 2.46% contribution of publications each. Following them, the institutes, 'University of Central Florida' in the USA (2.11%), 'McGill University'

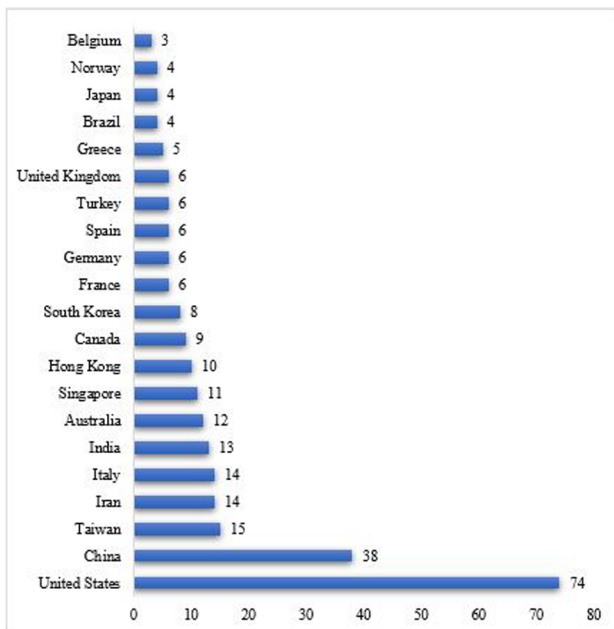


Fig. 8. No. of published articles across countries/regions (max. 3 number of publications).

(1.76%) in Canada, 'Purdue University' in the USA (1.76%), 'China University of Technology' (1.76%) have shown their respective research outputs in terms of the percentage of research contribution.

3.1.7. Distribution of articles by countries/regions

The contributions of different countries/regions in this domain have been analysed. The results from our investigations are depicted in Fig. 8. In this figure, the contributions (max. three number of publications) of different countries/regions are shown. The USA is ranked top in the list with 74 number of publications. China makes its position second in the list with 38 number of publications. Following them, Taiwan (15), Iran (14), Italy (14), India (13), Australia (12) have shown their respective research outputs in terms of the number of publications. Some countries, namely Brazil, Japan, Norway, and Belgium have published very few papers in this domain.

It is noteworthy to mention that the distribution of articles by countries/regions can be different from the results of the distribution by author's countries/regions or institutes' countries/regions because the count depends on their respective occurrences in papers. For example, consider Fig. 7 and Fig. 8. In Fig. 8, the total number of publications in the United States (74) is higher than that of India (13), but Fig. 7 indicates the % contribution of IIT Kharagpur (3.17%) in India is the highest. It is possible because the total publications made by all the institutes in the USA are the highest, whereas, according to the institute-wise contribution, the institute 'IIT Kharagpur' in India contributes majorly than any institute in the USA.

3.1.8. Distribution of articles by sponsoring agencies

The contributions of sponsoring agencies have been investigated with a maximum of two number of publications. The results are depicted in Fig. 9. It is revealed that the 'National Natural Science Foundation of China' sponsor the maximum number of research with 20 counts. It is followed by 'National Institute for Occupational Safety and Health' of USA (6 counts), 'National Science Foundation' of USA (5 counts), 'Ministry of Human Resource Development' of India (4 counts), 'Ministry of Steel, Government of India' (4 counts), 'National Science Council' of Taiwan (4 counts), and 'Tata Steel' of India (4 counts).

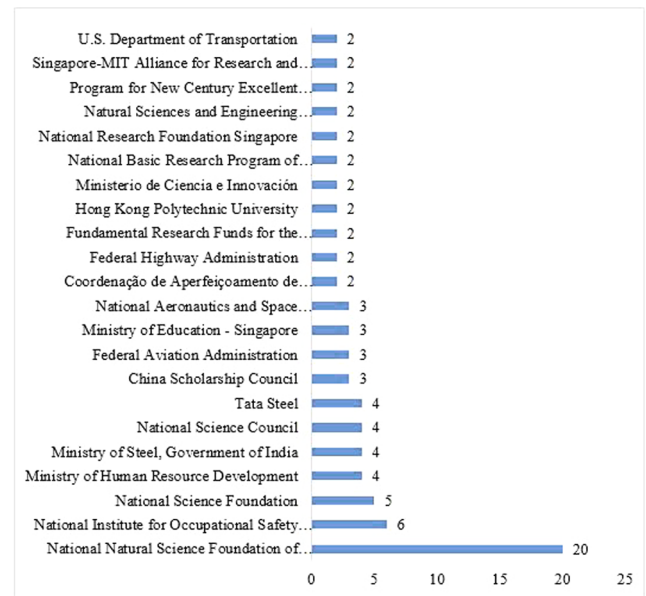


Fig. 9. No. of published articles with different sponsoring agencies (max. 2 number of publications).

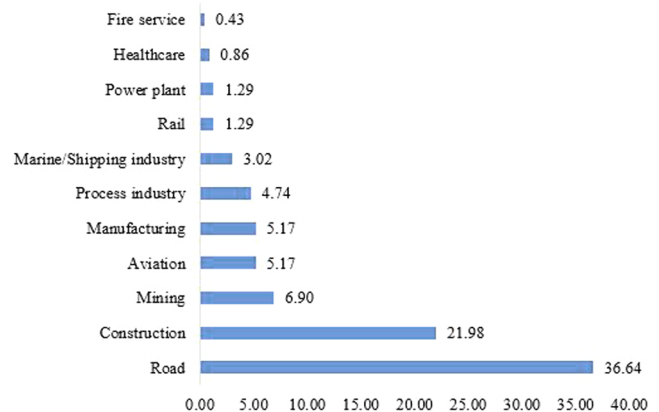


Fig. 10. The % contributions of different industries/sectors/domains.

3.1.9. Distribution of empirical papers by industries/sectors/domains

We investigated the studies in terms of the use of industry or sectors or domains. In Fig. 10, the percentage of contributions from different domains are recorded. From this figure, it is evident that the most of the researches focus on road accidents (36.64%), which is followed by construction (21.98%), mining (6.90%), aviation (5.17%), and manufacturing (5.17%). A few studies target on fire service (0.43%), and healthcare (0.86%) sectors.

3.2. Scientometric analysis

In the Scientometric analysis, the focus is given on the journal sources, authors, authors' keywords, documents/articles, and countries/regions to identify their influences in the domain of ML-based occupational accident analysis.

3.2.1. Analysis using journal sources

Using VOSViewer, the minimum citations and minimum number of articles of a journal are set at 1 and 1, respectively. It filters 58 out of 64 journals meeting the thresholds. The number of journals in the largest connected journal set is 34. The result is displayed in Fig. 11. According to the node and size of the fonts, it is evident from this figure that 'Safety Science' is the most productive journal as it has been publishing

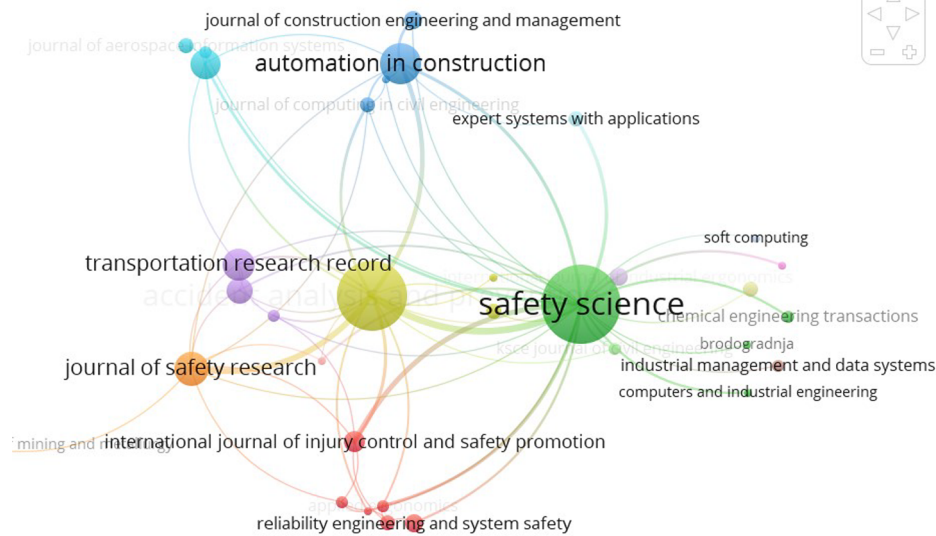


Fig. 11. Visualization of journal sources.

Table 1

A summary explaining the impacts of top journals in ML.

Journal	Number of publications	Total citation	Normalized citation	Average publication year	Average citation	Average normalized citation
Safety Science	44	777	54.84	2014	17.66	1.25
Accident Analysis and Prevention	36	1036	43.20	2014	28.78	1.20
Automation in Construction	15	282	20.65	2015	18.80	1.38
Transportation Research Part C: Emerging Technologies	9	335	18.79	2014	37.22	2.09
Journal of Safety Research	11	588	12.92	2009	53.45	1.17
IEEE Transactions on Intelligent Transportation Systems	7	201	11.57	2016	28.71	1.65
Reliability Engineering and System Safety	4	55	5.03	2015	13.75	1.26
International Journal of Industrial Ergonomics	4	19	4.14	2012	4.75	1.03
Journal of Computing in Civil Engineering	3	48	3.57	2015	16.00	1.19
Soft Computing	1	3	3.22	2019	3.00	3.22
International Journal of Injury Control and Safety Promotion	5	14	3.15	2017	2.80	0.63
Risk Analysis	3	138	2.86	2009	46.00	0.95
Ergonomics	2	71	2.71	1999	35.50	1.36
Decision Support Systems	3	20	2.63	2015	6.67	0.88
Applied Ergonomics	2	42	2.29	2015	21.00	1.15
International Journal of Project Management	1	89	2.26	2010	89.00	2.26
Expert Systems with Applications	3	53	2.01	2012	17.67	0.67
Journal of Aerospace Information Systems	3	27	1.60	2016	9.00	0.53
Computers and Industrial Engineering	1	1	1.07	2019	1.00	1.07
Computers in Industry	2	15	1.03	2007	7.50	0.52
Journal of Loss Prevention in The Process Industries	1	21	0.80	2013	21.00	0.80
European Transport Research Review	1	8	0.63	2017	8.00	0.63

the maximum number of publications. It is followed by ‘Accident Analysis & Prevention’, ‘Automation in Construction’, and ‘Journal of Safety Research’. The colours and the connecting lines represent the interrelationships among the journals. It shows that the articles of one journal cite other articles of different journals. For instance, these journals, namely ‘Safety Science’, ‘Accident Analysis & Prevention’, ‘Automation in Construction’, and ‘Journal of Safety Research’ have been actively citing each other.

A detailed quantitative representation of the influence or impact of journals is given in Table 1¹. In this table, all the journals are listed

according to their normalized citations. Based on this, the journals ‘Safety Science’, ‘Accident Analysis and Prevention’, and ‘Automation in Construction’ top the list. According to the average normalized citations, it is explored that the journals, namely ‘Soft computing’,

(footnote continued)

published in the same year and included in the data that is provided to VOSviewer. The normalization corrects for the fact that older documents have had more time to receive citations than more recent documents. Similarly, the normalized citations of a journal, author, keyword, and country have been calculated (van Eck and Waltman, 2009). The average normalized citations received by the documents in which a keyword or a term occurs or the average normalized citations received by the documents published by a source, an author, an organization, or a country (van Eck and Waltman, 2009).

¹ Note: The normalized citations of a document equal the number of citations of the document divided by the average number of citations of all documents

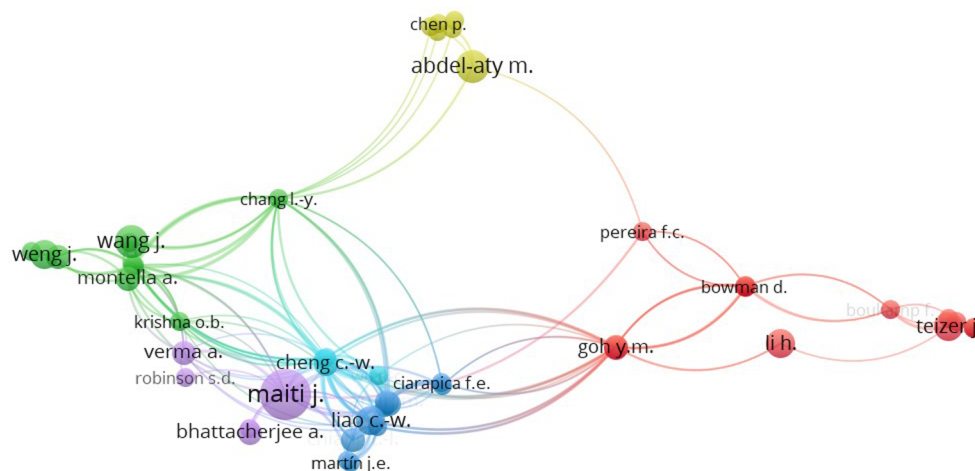


Fig. 12. Visualization of authors.

Table 2

A summary exploring the impacts of authors.

Authors	Number of articles	Total citation	Normalized citation	Average publication year	Average citation	Average normalized citation
Maiti J.	10	121	8.86	2012	12.1	0.89
Abdel-aty M.	5	186	8.02	2015	37.2	1.6
Teizer J.	4	148	7.4	2016	37	1.85
Wang J.	5	155	6.54	2014	31	1.31
Montella A.	3	150	5.46	2011	50	1.82
Verma A.	3	42	5.35	2017	14	1.78
Boukamp F.	2	131	5.23	2013	65.5	2.61
Cheng C.-W.	3	145	4.86	2012	48.33	1.62
Chang L.-Y.	2	370	4.68	2006	185	2.34
Leu S.-S.	2	124	4.06	2011	62	2.03
Lin C.-C.	2	124	4.06	2011	62	2.03
Miranda-Moreno L.F.	2	91	3.75	2015	45.5	1.88
Aria M.	2	93	3.72	2012	46.5	1.86
D'ambrosio A.	2	93	3.72	2012	46.5	1.86
Mauriello F.	2	93	3.72	2012	46.5	1.86
Yan X.	3	36	3.61	2017	12	1.2
Goh Y.M.	3	12	3.6	2018	4	1.2
Ubeynarayana C.U.	3	12	3.6	2018	4	1.2
Bowman D.	2	49	3.56	2017	24.5	1.78

'International Journal of Project Management', and 'Transportation Research Part C: Emerging Technologies' are ranked top of the list. It denotes that they are influential journals in ML-based occupational accident analysis domain in terms of the average normalized citation; whereas, based on the total citations and the total number of articles, 'Safety Science' and 'Accident Analysis and Prevention', 'Automation in Construction', and 'Journal of Safety Research' are found to be the four top-most productive journals, which have been publishing articles in this domain. The 'average publication year' indicates the average publication year of all the papers which are published in the similar sources of a journal. It helps to measure the recentness of the publications from a given journal. It is noteworthy to mention that the recentness can vary somewhat significantly. The journals like 'Soft Computing', 'Computers and Industrial Engineering', 'European Transport Research Review' and 'International Journal of Injury Control and Safety Promotion' have been publishing articles more actively in this domain. In contrast, journals like 'Ergonomics', 'Risk Analysis', and 'Computers in Industry' do not appear to be active in publishing articles in this domain.

3.2.2. Analysis using authors

Using VOSViewer, we first set the minimum citations and minimum number of articles of an author at 12 and 2, respectively. It filters 79 out of 673 authors meeting the thresholds. The number of the authors in the largest connected author set is 50, which is displayed in Fig. 12. The circle and font size for each author denote the quantity of publications in this domain. The authors displayed in Fig. 12 are clustered into six groups based on their own citation networks. For instance, 'Cheng C.-W.', 'Leu S.-S.', 'Lin C.-C.', and 'Wu T.-C.' appear in the similar cluster which indicates the mutual influence offered by collaborative citations. The influence/impact among authors has been further assessed with the help of the connection lines and distances, as shown in Fig. 12. For instance, 'Liao C.W.', 'Martin J.E.', and 'Ciarapica F.E.' can be found with strong linkage in this domain. The impact of authors is listed in Table 2², where all the authors are listed based on their 'Normalized

² Note: The full names of authors used in this table are as follows: Maiti J. = Maiti, Jhareswar; Abdel-aty M. = Abdel-Aty, Mohamed A.; Teizer J. = Teizer, Jochen; Wang J. = Wang, Jinxian; Montella A. = Montella, Alfonso; Verma A. = Verma, Abhishek; Boukamp F. = Boukamp, Frank; Cheng C.-W. =

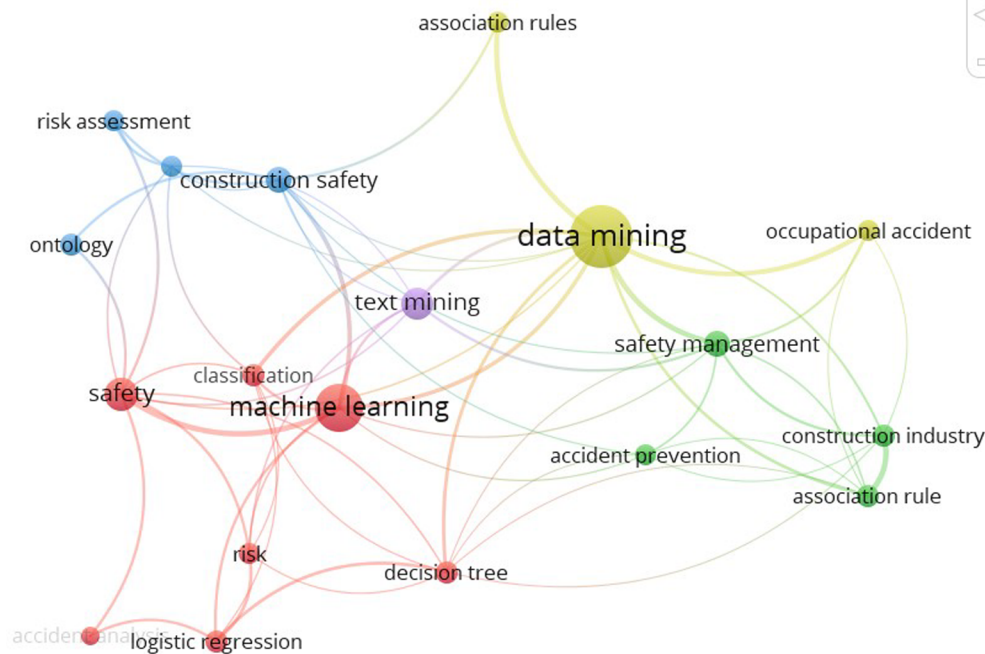


Fig. 13. Visualization of author keywords.

Table 3
Quantitative summary of the influence of keywords.

Keyword	Occurrence	Average publication year	Average citation	Average normalized citation
text mining	12	2017	16.75	1.77
risk assessment	6	2013	19	1.3
safety	9	2012	46.67	1.27
management				
association rules	6	2015	23.67	1.26
ontology	7	2014	25.29	1.12
construction	7	2012	32.43	1.09
industry				
data mining	37	2013	31.11	1.02
decision tree	7	2013	19.14	0.95
machine learning	24	2015	8.42	0.95
accident analysis	5	2013	19.4	0.77
logistic regression	7	2009	18.71	0.69
classification	7	2014	14.43	0.56
support vector machine	6	2016	11	0.49

Citation' score. According to this score, the authors 'Maiti J.', 'Abdel-aty M', and 'Teizer J.' are ranked at the top of the table. According to the average normalized citation score, it is evident that although 'Boukamp F.' has only two number of publications, he has been identified as an author with the maximum influence as measured by the 'Average Normalized Citation'. According to the total citation and average citation, the authors 'Chang L.-Y.', 'Abdel-aty M.', and 'Wang J.' are considered to have the highest contribution in this research domain. The authors, 'Maiti J.', 'Abdel-aty M.', 'Wang J.', 'Teizer J.', 'Liao C.-W.', and 'Weng J.' can be considered as the most productive authors based on

(footnote continued)

Cheng, Chingwu; Chang L.-Y. = Chang, Li Yen; Leu S.-S. = Leu, Sousen; Lin C.-C. = Lin, Chenchung; Miranda-Moreno L.F. = Miranda-Moreno, Luis F.; Aria M. = Aria, Massimo; D'ambrosio A. = D'Ambrosio, Antonio; Mauriello F. = Mauriello, Filomena; Yan X. = Yan, Xuedong; Goh Y.M. = Goh, Yangmiang; Ubeynarayana C.U. = Ubeynarayana, Chalani Udhayami; Bowman D. = Bowman, Dean.

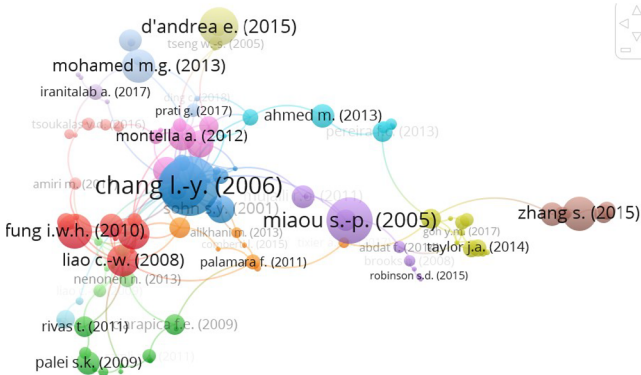


Fig. 14. Visualization of documents.

their number of publications in the body of literature. As per the 'Average publication year', most of the authors remain active in this domain of research, except 'Perng Y.-H.', 'Ciarapica F.E.', 'Giacchetta G.', 'Chang L.-Y.', and 'Bhattacharjee A.' as their average publication year values lie before 2010.

3.2.3. Analysis using authors' keywords

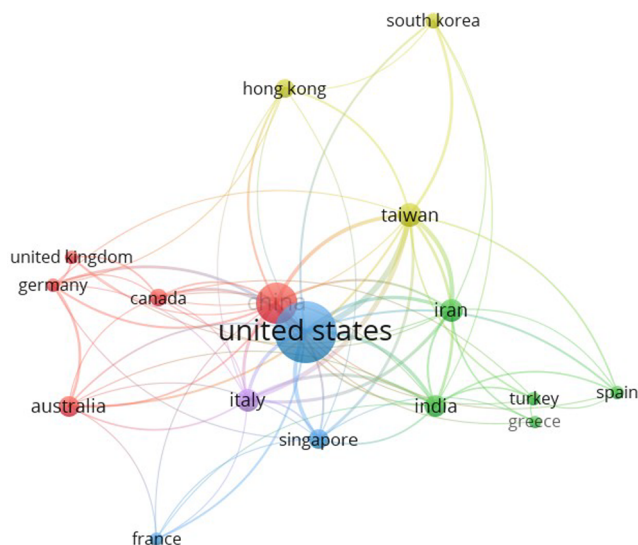
Keywords denote the main contents of any study and narrate research themes in a particular domain. Keywords co-occurrence represents the inter-closeness relationships among them. Initially, the general keywords are removed, for example, 'aviation safety', 'safety', 'manufacturing site', and others. Thereafter, the keywords having the same meaning semantically, are combined, for example, 'SMS' and 'Safety Management System', 'RF', and 'random forest', 'ML' and 'machine learning', etc. Then, using VOSViewer with fractional counting³,

³ Note: Fractional counting is a method used to generate a map based on bibliographic data. There is also another method available for this purpose, which is called a full counting method. The main aim of fractional counting is to minimize the influence of documents with many authors. During fractional counting, the strength of a co-authorship link between two authors is determined by two factors: (i) the number of documents co-authored by the

Table 4

A summary showing the articles with higher citations.

Article	Total citation	Normalized citation
Shao et al. (2019)	6	6.44
Comberti et al. (2018)	7	4.27
Zhang et al. (2015)	86	3.85
Kuhn (2018)	6	3.66
Jana et al. (2019)	3	3.22
Mohamed et al. (2013)	82	3.12
Poh et al. (2018)	5	3.05
Ding et al. (2018)	5	3.05
Chang and Wang (2006)	214	3
Montella et al. (2012)	61	2.74
Tixier et al. (2016b)	36	2.54
Verma et al. (2014)	38	2.44
Fung et al. (2010)	89	2.26
Liao and Perng (2008)	76	2.15

**Fig. 15.** Visualization of countries/regions.

we set the minimum number of occurrence of a keyword as 5, which results in the selection of the number of keywords as high as 19 out of 750 (Fig. 13). The keywords, including ‘association rules’, ‘classification’, ‘decision tree’, ‘logistic regression’, and ‘machine learning’ have been used more frequently in this domain.

Keywords displayed in Fig. 13 are clustered into five groups. Keywords in the same cluster have closer inter-relationships. For instance, association rules are often co-studied in construction safety. The distance and connection line between keywords denote the inter-relatedness among them. For instance, support vector machine and ontology are identified to have a closer relationship with risk management in construction safety. The font size denotes the count of the keyword in the collected articles. The keywords studied more frequently include ‘accident analysis’, ‘machine learning’, ‘risk assessment’, ‘text mining’.

The influence of these keywords is presented in Table 3, where the keywords are ranked according to the ‘average normalized citation’. The keywords ‘text mining’, ‘risk assessment’, ‘association rules’, ‘safety

management’, ‘ontology’ are ranked top according to the average normalized citation score. It implies the fact that the use of text mining or risk assessment or rule extraction is gaining more importance in this domain of research. The ‘average publication year’ explores the recentness of the keywords being used in this research community. It is observed that most of the keywords in Table 3, including ‘text mining’, ‘support vector machine’, ‘association rules’, ‘ontology’, and ‘classification’ are gaining more importance in recent years except the use of ‘logistic regression’. According to the number of occurrences, the keywords, including ‘text mining’, ‘ontology’, ‘classification’, and ‘decision tree’ are placed at the top of the table.

3.2.4. Analysis using articles

For analysing Scientometric analysis using articles or documents, first, we use fractional counting method and set the minimum number of citation of a document as 1. As a result, the number of documents selected is 195 out of 232 after meeting this threshold. Out of 195 documents, a total of 112 documents form the largest set of a connected network, which is visualized in Fig. 14. The articles identified by high citations are summarised in Table 4. The most influential articles based on the normalized citation are shown in this table. It is to be noted that not all the 112 articles are listed here, but only the top few articles with their respective normalized citation. For instance, the study of Shao et al. (2019) focused on the investigation of pattern extraction of accidents in a construction project in China, Comberti et al. (2018) used a large database for accident analysis, Zhang et al. (2015) used ontology in semantic modeling for maintaining safety in construction. From these articles, it is evident that in the application of ML in occupational accident/incident domain, ontology, text mining, classification, and rule extraction are gaining more attention from the research community.

3.2.5. Analysis using countries/regions

Using VOSViewer, we use fractional counting method and set the maximum number of authors per document as 25, the minimum number of documents of a country/region as 5, and the minimum number of citations of a country/region as 0. Using these thresholds, the number of countries/regions is selected 17 out of 43, which also forms the largest set of a connected set of countries/regions. This set is displayed in Fig. 15. The circle size and font for each country/region denote their quantity of publications in this domain. The countries/regions displayed in this figure are clubbed into five clusters based on their citation networks. To exemplify, ‘Australia’, ‘Canada’, ‘China’, ‘Germany’, and ‘United Kingdom’ appear in the similar cluster which indicates the mutual influence/impact by giving citations to each other’s study.

The impact within countries/regions can further be computed by the connection lines and distance, as shown in Fig. 15. For instance, ‘Australia’, ‘Canada’, ‘China’, ‘Germany’, and ‘United Kingdom’ can be found with a strong linkage in this domain. The influence of research of countries/regions is provided in Table 5, where all the countries/regions are listed based on their ‘Normalized Citation’ score. Based on this score, the countries/regions, namely ‘The United States’, ‘China’, ‘Italy’, ‘Taiwan’, and ‘Australia’ top the table. According to the ‘Number of citations’, ‘The United States’, ‘Taiwan’, ‘Italy’, ‘China’, and ‘Australia’ are ranked at the top and they are considered to have the highest contribution to the academic body of knowledge in this domain. It is also evident that ‘Australia’, ‘Canada’, ‘China’, ‘France’, and ‘Germany’ are the top five countries in the table according to their ‘Average normalized citation’ score. As per the ‘Average publication year’, most of the countries/regions remain active in this domain of research, except ‘The United Kingdom’, ‘France’, and ‘Greece’ as the value of their average publication year lie before the year 2010.

3.3. Citation network analysis

To classify the overall research in this domain, we have used

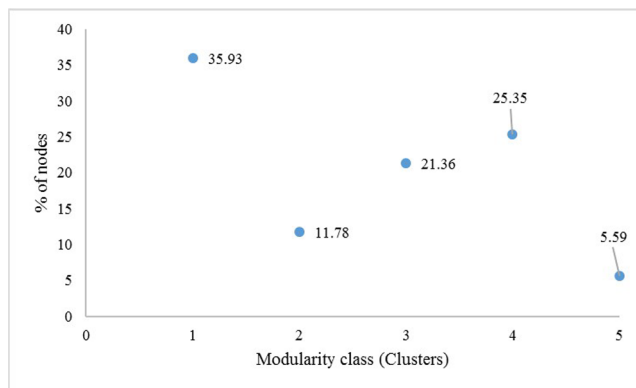
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authors, and (ii) the total number of authors of each of those documents. For fractional counting, when an author has co-authored a document with n number of authors, the strength of each of the n co-authorship links becomes $\frac{1}{n}$. The total strength of such links becomes equal to 1, which contrasts with the full counting method. In full counting, each of the links (of n co-authorship) has a strength equal to 1, which results in a total strength equal to n .

Table 5

Summary of countries/regions having higher publication impacts based on average normalized citations of articles.

Country/regions	Articles	Citations	Norm. citations	Avg. pub. year	Avg. citations	Avg. norm. citations
United States	74	1527	80.95	2015	20.64	1.09
China	38	344	41.50	2016	9.05	1.09
Italy	14	407	21.11	2014	29.07	1.51
Taiwan	15	733	16.93	2011	48.87	1.13
Australia	12	326	15.24	2014	27.17	1.27
India	13	123	12.31	2014	9.46	0.95
Singapore	11	159	11.96	2015	14.45	1.09
Canada	9	209	10.77	2013	23.22	1.20
Germany	6	173	9.37	2017	28.83	1.56
South Korea	8	130	7.74	2015	16.25	0.97
Hong Kong	10	206	7.65	2014	20.60	0.77
Spain	6	173	7.00	2014	28.83	1.17
Iran	14	95	6.96	2015	6.79	0.50
Greece	5	78	5.53	2009	15.60	1.11
United Kingdom	6	145	5.32	2010	24.17	0.89
Turkey	6	50	4.53	2017	8.33	0.76
France	6	47	4.02	2010	7.83	0.67

**Fig. 16.** Modularity vs. percentage of nodes.

network modularity-based clustering approach (Newman and Girvan, 2004). This approach helps to detect the structure of the community of citation networks in the past literature. To implement this approach, Gephi software is used. Using this approach, a total of five clusters are obtained. The percentage of nodes in each of the clusters, i.e., modularity classes is shown in Fig. 16. The clusters of all the documents are depicted in Fig. 17, where different clusters are characterized by different colours. Further, investigating those clusters reveals some important topics or theme in ML-based occupational incident analysis. Most of the papers in the pink-coloured cluster indicate that the research in those papers relates to 'Incident prediction/Incident pattern extraction'. Similarly, the light green-coloured cluster indicates research on 'Risk prediction/Risk pattern analysis', the cyan-coloured cluster denotes research on 'Injury severity prediction/Severity pattern extraction', the red-coloured cluster indicates research on 'Incident outcomes prediction' and the deep green-coloured cluster indicates diverse research topics with small numbers, which we call as 'Others'. In the 'Others' group, there are different studies reported which explore the some of the fragmented and limited use of decision support system (DSS), big data or internet of things (IoT), image or video processing-based incident analysis and prevention.

3.4. Research growth

In this section, five research domains identified by the CNA are discussed. The development or growth of research on these five domains is identified. In addition, the growth of ML-based approaches on these domains is also mentioned.

3.4.1. Growth of research in research domains

(a) *Incident outcomes prediction*: In the area of incident prediction, there are many ML algorithms used, including SVM, ANN, ELM, and decision tree. They have been used in different application areas, including construction (Rivas et al., 2011), mining (Rivas et al., 2011), ship-building (Fragiadakis et al., 2014), and service (Matias et al., 2008).

To exemplify, Matias et al. (2008) used ELM, SVM, BN methods for predicting the incidents like floor-level falls. They used 148 samples collected from a number of companies from 2003 to 2006 in Spain. From their experiments, BN was identified to have a higher prediction than others. Sánchez et al. (2011) conducted one study using SVM to classify the workers suffering from work-related incidents for a year. They analysed the data consisting of 11,054 response samples of the workers employed in all economic activities in Spain. Their findings show that SVM performed better than back-propagation neural network (BPNNs). In 2011, Rivas et al. (2011) modeled the incidents in two companies in the construction and mining sectors to identify the most important causes of incidents and developed predictive models using BN, SVM, and other ML techniques. BN-based prediction models have also been used by many researchers in different sectors, such as mining (Matias et al., 2008), construction (Matias et al., 2008). For example, Sanmiquel et al. (2015) carried out one study in the Spanish mining sector and analysed the 69,869 instances of occupational accidents from 2003 to 2012 using BN. Another important prediction model used in this domain is ANN. Due to its important characteristics like the ability to learn from data, distributed memory, parallel operation and fault tolerance, it has been widely used in diverse fields of study, including occupational accident analysis domain. For instances, He et al. (2010) attempted to solve the problem of coal and gas outburst by classification technique using the backward algorithm of ANN (BA-ANN) and exponent evaluation method (EEM). Using BA-ANN, the weights of factors are calculated towards the response variables (i.e., the levels of coal and gas outburst). Yi et al. (2016) developed an early warning system for workers in hot and humid environments using ANN. They have collected 550 data related to work, environment, and individuals which are analysed by ANN to predict the rating of perceived exertion (RPE) of the workers in the construction sites.

Apart from the application of ANN in occupational accidents, there are several studies available for other accidents of which research on road accidents is found to have attained more focus (Alikhani et al., 2013). More interestingly, artificial intelligence (AI) approaches like ANN are found to have greater performance in terms of prediction than regression analysis. Reviewing the literature on accident analysis domain, techniques like SVM and ANN are found to be popular and useful as they have a robust theoretical grounding that enables the successful

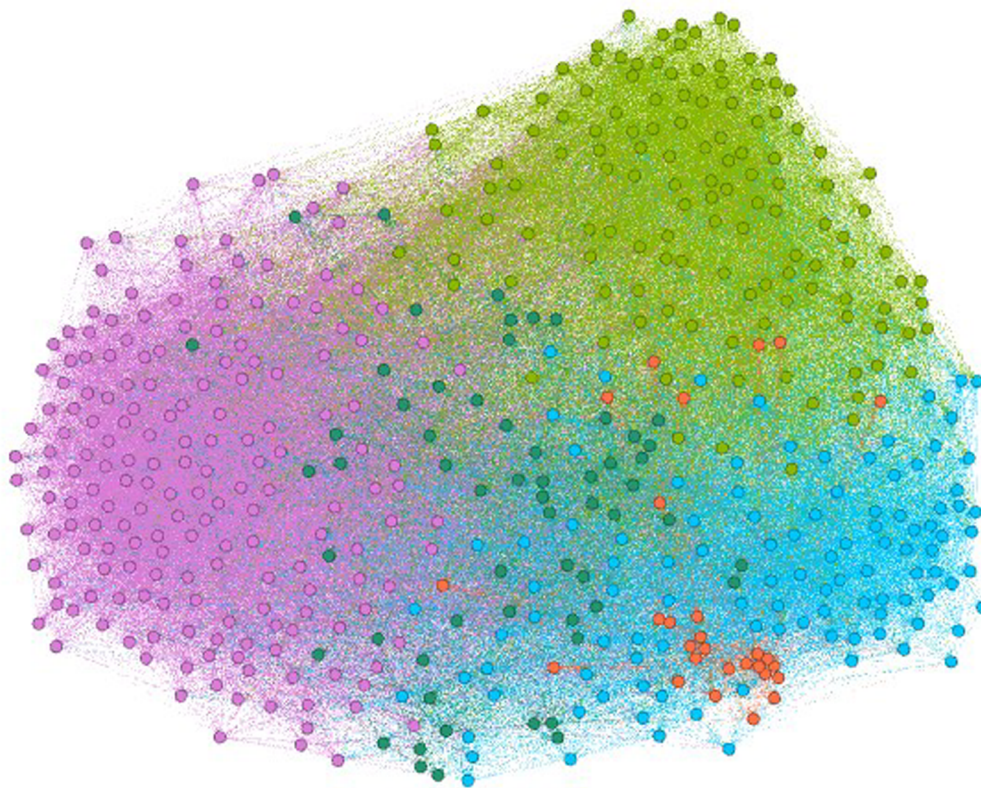


Fig. 17. Modularity-based five clusters of the selected papers.

learning from the data, capability of handling any level of complexity except computational complexity of the problem and flexibility with non-parametric philosophy. Goh and Ubeynarayana (2017) used SVM, LR, KNN, RF, NB, and DT algorithms to classify the construction accident narratives. From their experiments, SVM was found to be the best one with F1-score as high as 0.92. F1-score Wang et al. (2018) investigated the time and location of a secondary accident after the occurrence of an initial accident on the road. With the road accident data collected over three years in California, a shock wave-based method was introduced to identify secondary accidents. In this study, they have used linear regression, least squares support vector machine (LSSVM), and back-propagation neural network (BPNN). From their experiment, BPNN outperforms others in the prediction of a secondary accident. Ghodrati et al. (2018) developed a predictive model to predict the safety outcomes in the construction industry to help the organization make proactive strategic to curve the accident occurrences. In their experiment, multiple linear regression (MLR) was used which shows a satisfactory result in prediction. Sakhakarmi et al. (2018) proposed an ML-based classification algorithm, SVM to classify scaffolding failure cases and reliably predict safety conditions at a large, complex construction site. Their proposed approach produced prediction accuracy as high as 96%. In 2019, Ayhan and Tokdemir (2019) used MLR, ANN, fuzzy set theory to predict safety outcomes in the construction site. Fuzzy-based ANN algorithm was able to predict safety outcomes with accuracy 84%. Some of the selected articles focusing on incident outcome prediction are listed in Table 6.

(b) *Rule-based pattern extraction of incidents:* Prediction analysis, as a standalone tool, may not serve the entire purpose of the accident analysis until a prescriptive analysis is not made for the interpretation of the accident causation. Rule-based pattern extraction and its interpretation are often deemed to be an effective approach. The rules can usually be extracted using either the decision tree or association rule mining (ARM) algorithm. In several previous studies on our research domain, the decision tree has been implemented for the extraction and

interpretation of rules more frequently than others, including ARM. In fact, the decision tree observed to be effective when the target function is discrete-valued, when it is describable by attribute-value pairs, or when the data sets are noisily trained. In the application of decision tree algorithms in accident analysis, algorithms, including CART (Bevilacqua et al., 2008), Chi-square automatic interaction detection (CHAID) (Matias et al., 2008), Exhaustive CHAID (Persona et al., 2006), and Quick, Unbiased and Efficient Statistical Tree (QUEST) (Hajakbari and Minaei-Bidgoli, 2014; Persona et al., 2006), C4.5 (Matias et al., 2008), C5.0 (Cheng et al., 2012; Hajakbari and Minaei-Bidgoli, 2014), and others are usually used for prediction of occupational accident. The main aim to use DT is to predict and interpret qualitative and quantitative patterns lying in data which leads to the exploration of hidden information. Due to its relaxation on assumptions on the distribution of attributes or independence of attributes, DTs have been successfully used in different fields, including medicine (Oztekin et al., 2011), social sciences (Olson et al., 2012), business management (Aviad and Roy, 2011), construction engineering and management (Leu and Chang, 2013), mining industry (Amiri et al., 2016; Cheng et al., 2012; Goh and Ubeynarayana, 2017), process industry (Bevilacqua et al., 2008), refinery industry (Bevilacqua et al., 2008), petrochemical industry (Cheng et al., 2013), aviation industry (Babič et al., 2015), and others. The main reason for their wide applicability is that they work without any dependency on assumption on attributes; neither do they necessitate any transformation of variables, rather they can handle both categorical and continuous variables simultaneously and are equally effective in both classification and regression (Ferreira and Vasilyev, 2015; Konda, 2010). In addition to the use of a single decision tree algorithm, a collection of the decision tree has also been used. It is called ensemble decision tree algorithm. Random forest (RF), one of the ensemble learning algorithms, had been effectively used with the enhanced predictive power (Brown, 2016; Ittoo et al., 2016; Rajpathak, 2013; Tanguy et al., 2016; Liew et al., 2014; Tixier et al., 2016a; Tixier et al., 2016b).

Table 6
Summary of some selected articles focusing on incident outcome prediction.

Author (year)	Objective	Domain	Methods used	Key findings
Matias et al. (2008)	To develop a machine learning methodology for floor-level fall detection	Mining, construction, and service	DT, SVM, BN, ELM	BN outperforms others.
He et al. (2010)	To develop a machine learning methodology for prediction of coal and gas outburst outcomes	Mining	ANN, EEM, BA-ANN- based EEM	BA-ANN-based EEM performs better than others.
Sánchez et al. (2011)	To predict the work-related accidents according to working conditions	Survey	SVM, ANN	SVM performs better than ANN
Rivas et al. (2011)	To predict workplace accidents	Mining and construction	DT, BN, SVM, logistic regression	BN, DT produce satisfactory results
Sanmiquel et al. (2015)	To predict the types of accidents	Mining	DT, Predictive Apriori	DT performs better and produces satisfactory results
Yi et al. (2016)	To develop an early-warning system for site work in hot and humid environments	Construction	ANN	ANN produces good results with RMSE value 0.857 in prediction
Goh and Ubeynarayana (2017)	To develop a prediction model using text classification	Construction	SVM, LR, RF, KNN, DT, NB	SVM performs better than others with F1-score as high as 0.92
Wang et al. (2018)	To model when and where a secondary accident occurs	Road	LR, BPNN, LSSVM	BPNN outperforms others
Ghodrati et al. (2018)	To predict safety outcomes in the construction industry	Construction	MLR	MLR produces satisfactory results in prediction
Sakhakarmi et al. (2018)	To predict the scaffolding failure case and safety condition	Construction	SVM	SVM produces prediction accuracy as high as 96%
Ayhan and Tokdemir (2019)	Predicting the outcome of construction incidents	Construction	MLR, ANN, Fuzzy set theory	Fuzzy-based ANN is able to produce 84% accuracy in prediction

Table 7
Summary of some selected articles focusing on rule-based pattern extraction.

Author (year)	Objective	Methods used	Domain	Key findings
Persona et al. (2006)	To identify the risk group and factors behind the accidents	CART	Construction	CART is able to identify and analyse the factors for risk areas at the workplace
Bevilacqua et al. (2008)	To identify the relationships among the factors behind the accidents	CART	Petrochemical industry	CART is proved to be very efficient in extracting the relationships among the factors behind the accidents
Matias et al. (2008)	To analyse the patterns behind occupational accidents	BN, DT, SVM, ELM	Construction	BN is found to be the best classifier in predicting accident and extracting important factors behind accidents
Cheng et al. (2012)	To analyse the causes behind occupational accidents	CART	Construction	The results revealed that the occurrence rules for falls and collapses are useful to predict the occurrence of occupational injuries.
Cheng et al. (2013)	To analyse the causes behind occupational accidents	CART	Petrochemical industry	CART is capable of identifying the root causes behind the accidents
Mistikoglu et al. (2015)	To construct decision trees and to extract rules that show the associations between the input and output variables (attributes) for roofer fall accidents	C5.0, CHAID	Construction	Decision tree classifiers were able to produce high prediction accuracy and extract rule-based patterns from data for explaining the factors behind the roofer fall accident.
Sanmiquel et al. (2015)	To predict the types of accidents	BN, DT, Predictive Apriori	Mining	DT performs better and produces satisfactory results in prediction and extraction of rules behind accidents
Brown (2016)	To identify construction rail accident causes	NLP, RF	Railway	RF was capable of identifying causes behind rail accidents
Tixier et al. (2016a)	To predict and analyze construction accident cases.	NLP, SGTB, RF	Construction	RF was capable of predicting and identifying causes behind construction accidents
Amiri et al. (2016)	To extract patterns of high-risk accidents	CART, RF, ARM	Construction	Proposed algorithms were capable of extracting important patterns from data that could explain high-risk accidents at work

Table 8
Summary of some selected articles focusing on injury risk prediction.

Author (year)	Objective	Methods used	Domain	Key findings
Palei and Das (2009)	To predict the risk of roof-fall in a coal mine	LR	Mining industry	The results revealed that wider gallery width, thin seams, unsupported or partially supported roofs, deep workings are more prone to major accidents.
Giarapica and Giachetta (2009)	To predict the occupational injury risk	ANFIS, ANN, Holt-Winters model, LR, Moving average	NA	ANFIS performs better than other algorithms with 7.76% MAD value in prediction.
Meng and Weng (2011)	To predict the work zone casualty risk	GA, LR	Road	GA outperforms LR in prediction accuracy. A set of 19 rules are identified. The seat position has the most significant effect on reducing the prediction error of the GA approach. The variables of the posted speed limit, traffic control devices and truck or bus involved are found to have no effects on improving the prediction performance.
Weng et al. (2013)	To predict the work zone casualty risk	DT, LR, DT-LR	Road	DT-based LR outperforms DT and LR in prediction.
Leu and Chang (2013)	To assess the safety risk in construction projects	BN	Construction industry	The model can accurately provide the probabilities of safety risks and identify the causes of accidents with their relationship.
Hajakbari and Minaei-Bidgoli (2014)	To assess the occupational accident risk	k-means, CHAID, CART, C5.0, QUEST	Overall database (labour data)	The results show that the proposed method could identify and evaluate critical points with a high level of accuracy.
Fragiadakis et al. (2014)	To predict the occupational risk in a shipbuilding industry	ANFIS	Shipbuilding/ Marine industry	The proposed model has the ability to predict the risk with the average error of prediction 6.1% when the Gaussian type membership function is used.
Tsoukalas and Fragiadakis (2016)	To predict the occupational risk in a shipbuilding industry	MVLR, GA, ANFIS	Shipbuilding/Marine industry	GA-based MVLR performs better than ANFIS in risk prediction.
Debnath et al. (2016)	To assess the occupational risk in construction	ANFIS	Construction industry	ANFIS performs satisfactorily in prediction and rule generation.
Amiri et al. (2017)	To assess the hazards in construction based on risk	A fuzzy probabilistic rule-based approach	Construction industry	The model was capable to rank the risks of different types.
Kaeeni et al. (2018)	To predict the risk of derailment accident	ANN, DT, NB, SAVG, MVOTE, and ECGA	Railway	ECGA outperforms others with 88.45% prediction accuracy.
Jana et al. (2019)	To predict the occupational risk	IT2FLC-based ANFIS, MLR	Construction industry	IT2FLC-based ANFIS performs the best with 5.18% RMSE value in prediction.
Jahangiri et al. (2019)	To predict the risk of falling from a scaffold	LR, ANFIS	Construction industry	ANFIS produces better prediction performance (MAPE 1.173) than LR. The analyses indicate that platform, joints, ladders, personal protective equipment and guardrails are the most important factors.

Note: ECGA-Ensemble classifiers-based GA; SAVG-Simple AVeraGe; MVOTE: Majority VOTE; IT2FLC: Interval type-2 fuzzy logic controller.

To exemplify, [Persona et al. \(2006\)](#) carried out an analysis of occupational injury data of 156 cases collected from 2000 to 2002. They have used the CART algorithm for the analysis. From their experiments, the causes behind the work-related injuries were identified. The use of CART is found to be a very effective model for identifying potential risk areas in the workplace. [Matias et al. \(2008\)](#) proposed a methodology to predict the floor-level falls and analysis of their causes in the construction industry. A set of ML-based algorithms, namely BN, CART, SVM, and ELM are used in this study. BN is found to be effective in their experiments based on its predictive and interpretative capabilities. [Bevilacqua et al. \(2008\)](#) used CART algorithm for identification of the important relationships among the attributes or factors responsible for the occurrence of incidents in a medium-sized refinery. Using their approach, the cause-effect relationships are identified between the factors and the accidents. As a consequence, the influence of various factors on the high-risk situation is obtained. [Cheng et al. \(2012\)](#) used CART decision tree algorithm on 1542 accident data collected during 2000 to 2009 from the construction industry to identify the cause-effect relationships responsible behind occupational incidents in the plant. The rules obtained from CART helped both the private and public construction industries to predict the occupational injuries. The derived rules are found to be potential to effectively improve the safety training and practices for workers. [Mistikoglu et al. \(2015\)](#) used C5.0 and CHAID decision tree algorithms in the construction industry to predict and analyse the factors behind the occurrences of roofer fall accidents. [Brown \(2015\)](#) used NLP to identify the contributors to rail accidents and implemented RF to check the predictive power of the contributors. [Tixier et al. \(2016a\)](#) used natural language processing (NLP) to identify the precursors of accident from accident narratives. Later, they implemented two ensemble techniques, namely RF and stochastic gradient tree boosting (SGTB) to predict of four response variables, namely energy source, types of injury, body parts injured, and severity of the injury. Some of the selected articles focusing on rule-based pattern extraction are listed in [Table 7](#).

(c) *Injury risk prediction:* Prediction of risk of injury at workplaces emerges as a focused area of research and practice. The people at work should be exposed to the minimum acceptable level of risk and most of the hazards at the workplace should be eliminated or controlled by prevention through design (PtD). By definition, the risk is a measure of hazard potential, where the hazard can be defined as a condition or situation that can result in harm to people, property, and environment ([Fragiadakis et al., 2014](#)). The paths from hazards to accidents are seldom visible and hence, proper modeling is required to trace out or predict the paths so that effective preventive measures can be put into the system much before an accident could take place ([Babur et al., 2016](#)). A number of earlier studies concentrated on risk analysis in different domains; for instances, in shipbuilding industry ([Pereira et al., 2013](#); [Tsoukalas and Fragiadakis, 2016](#)), process industry ([Alileche et al., 2017](#)), road ([Alikhani et al., 2013](#)), and so forth.

There are several techniques used for predicting occupational risk; some of them are parametric techniques, such as negative binomial regression, Poisson regression, or logistic regression, and some of them are non-parametric ML techniques, such as ANN, SVM, DT, and others. For examples, [Palei and Das \(2009\)](#) used linear regression to predict the risk of roof-fall accident in a coal mine. They have used 128 roof fall accident data collected from five underground coal mines in India. Experimental results revealed that LR is capable of producing high accuracy in prediction of accident risk. Some factors, including wider gallery width, thin seams, deep workings were found to be more hazardous. [Meng and Weng \(2011\)](#) used a genetic algorithm (GA) to assess the casualty risk at a work zone. Their approach outperformed binary logistic regression based on prediction accuracy. From their analysis, it was explored that the seat position effects significantly on the prediction of risk. [Weng et al. \(2013\)](#) used DT, LR, and DT-based LR for predicting work zone casualty risk at the road. Their proposed DT-based LR outperforms other algorithms in prediction. [Leu and Chang](#)

(2013) deployed BN also in the construction industry to assess the safety risk. From their study, it was explored that the algorithms can accurately provide the probabilities of risks and identify the factors leading to incidents. [Hajakbari and Minaei-Bidgoli \(2014\)](#) used *k*-means clustering, CHAID, CART, C5.0, and QUEST on a large overall labor database to identify and evaluate critical factors for accidents.

The parametric models with their inherent distributional assumptions (i.e., follows a particular probability distribution) fail to extract the hidden information and knowledge (e.g., patterns or rules) from this type of complex datasets. The non-parametric ML techniques overcome the above issues as the ML algorithms, such as ANN can learn and model non-linear relationships. They can generalize and infer hidden relationships from both structured and unstructured data. However, one of the drawbacks of ANN is its inability to capture the uncertainty in risk-based analyses because of the existence of bias towards the probability of occurrence of accidents, inadequate collection of data, and lack of generalizability ([Hallowell et al., 2011](#)). The application of fuzzy rules is a solution to this problem. Therefore, integrating a neural network with fuzzy rules can be an effective approach, which not only takes care of the complex and non-linear relationships among the variables but also takes fuzziness of data into account ([Fragiadakis et al., 2014](#)). Adaptive neuro-fuzzy inference system (ANFIS), is an ML technique, which is capable of handling both the above-mentioned issues. Due to its network architecture, it is more capable of predicting risk and is used in diverse fields of studies, for examples, coal mine ([Adoko et al., 2013](#)), road ([Alikhani et al., 2013](#)), railway ([Ćirović and Pamučar, 2013](#)), shipbuilding ([Fragiadakis et al., 2014](#)), inspection ([Rachman and Ratnayake, 2019](#)), electric power industry ([Marhavilas and Koulouriotis, 2012](#)), nuclear power industry ([Lee and Lee, 2006](#)), and others. Moreover, the use of ANFIS is very effective in prediction than conventional ANN in some of the earlier studies ([Alikhani et al., 2013](#); [Bakirtzis et al., 1995](#)). Besides, reports by [Bakirtzis et al. \(1995\)](#) explored the supremacy of ANFIS over the traditional ANN.

To exemplify, [Adoko et al. \(2013\)](#) developed the ANFIS model to predict rock-burst intensity using 174 coal mine data. The model was tested with five other models, namely Fuzzy C-Mean-based Takagi–Sugeno Fuzzy Inference System, Subtractive clustering-based Takagi–Sugeno Fuzzy Inference System, Fuzzy C-Mean clustering-based Mamdani Fuzzy Inference System, knowledge-based Mamdani Fuzzy Inference System, and ANFIS. The experimental results revealed a superior performance of the ANFIS model over other algorithms in terms of prediction accuracy. [Fragiadakis et al. \(2014\)](#) used the ANFIS model to predict occupational risk and analysed the effect of working conditions on injury in the shipbuilding industry. From their experiment, ANFIS was found to predict occupational risk more accurately. Their proposed model was able to predict the risk with the average error of prediction of 6.1%. In addition, many factors behind the risk are also identified. Further, [Tsoukalas and Fragiadakis \(2016\)](#) developed a GA-based multivariate linear regression (MVLr) to predict occupational risk. The results explored that the hybridization of GA with MVLr gives better prediction accuracy. [Debnath et al. \(2016\)](#) used the ANFIS model to predict occupational risk in the construction industry. Their proposed approach performed satisfactorily in prediction and the rule generation. [Amiri et al. \(2017\)](#) used a fuzzy probabilistic rule-based approach to rank the risk of different types of accidents in a construction area. Later, in the construction industry, [Jana et al. \(2019\)](#) have used interval type-2 fuzzy logic controller-based ANFIS model to predict the occupational risk, [Jahangiri et al. \(2019\)](#) used LR and ANFIS models to predict the falling risk from a scaffold, which produces a better prediction performance than LR and also identifies some important factors responsible for higher risk. Some of the selected articles focusing on injury risk prediction are listed in [Table 8](#).

(d) *Injury severity prediction:* To augment the degree of safety, risk-based analyses have been a popular choice among the researchers. A number of practices have emerged for this purpose for modeling risk of workplaces. The injury risk is usually quantified by multiplying the

Table 9
Summary of some selected articles focusing on injury severity prediction.

Author (Year)	Objective	Methods used	Domain	Key findings
Chang and Chien (2013)	To predict and analyze injury severity of truck accidents	CART	Road	The results revealed that vehicle type, drinking-driving, collision type, seat belt use, and driver/vehicle action, number of vehicles involved in the accident and accident location were the key determinants of injury severity for truck accidents.
Delen et al. (2013)	To predict the injury severity of drivers	ANN, SVM, DT, LR	Road	The relationships between different levels of injury severity and the crash-related risk factors are identified.
Esmaili et al. (2015)	To predict the injury severity of construction workers	GLM	Construction industry	The findings of their research explored that severity level could be predicted by the elements of the working environment, such as equipment, tools, activities, physical attributes, and more importantly, environmental conditions of the workplace.
Chen and Luo (2016)	To predict injury severity of workers from the incident 'falling from height'	k-modes clustering, radial basis function network (RBFN), and decision tree (DT)	Construction industry	The results indicate that the DT-based falling risk prediction model has the best performance of 75% accuracy and the top three critical factors of falling event's severity are the distance from the ground, worker's occupation and the source of the falling.
Tixier et al. (2016b)	To predict injury severity of construction workers	NLP, SGTB, RF	Construction industry	Injury severity is mainly random. Both RF and SGTB perform poor in injury severity prediction.

probability of any incident resulting in an injury that happened with its expected or predicted severity. Although the two factors are equally significant in risk quantification, the main focus of the earlier studies is often concentrated on the probability of the occurrence of an injury and their causes. The primary reason behind this trend could be the availability of the data which represent the physical attributes of any particular work environment leading to injury, while the latter one receives comparatively little attention since the nature of prediction of injury severity is random (Tixier et al., 2016a). Nevertheless, the prediction of injury severity is essential. According to Hallowell et al. (2011), society can reap the benefits from the capability to accurately predict the severity of a potential injury. With the availability of a proper predictive model, researchers could attempt to predict more accurately the level of injury severity, thereby assessing risk empirically for a particular work environment. Besides, the prediction of injury severity can be useful in the identification of near-miss situations of high impact (Hallowell et al., 2017). Realizing the need of this research, a few studies have been carried out for injury severity prediction in diverse applications, for instances, road accidents (Buendia et al., 2015; Chang and Chien, 2013; Chen et al., 2016a; Chen et al., 2016b; Chen and Luo, 2016; Delen et al., 2013), construction (Esmaili et al., 2015; Hallowell et al., 2017; Tixier et al., 2016a), etc.

To exemplify, Chang and Chien (2013) used CART models for predicting injury severity of drivers in traffic accidents. They also identified the contributing factors for the occurrence of road accidents. Delen et al. (2013) analysed the risk factors that contribute to the injury severity in motor vehicle crashes. Their proposed approach was able to capture the intricate relationships between different levels of injury severity and different risk factors. In 2016, Chen and Luo (2016) attempted to predict injury severity of workers from the incident 'falling from height' using k-modes clustering, radial basis function network (RBFN), and decision tree. The results denote that their proposed model for falling shows the superior performance of 75% accuracy. In addition, the three important factors of falling event's severity are identified as a worker's occupation, distance from the ground, and the source of the falling. Esmaili et al. (2015) used a generalized linear model (GLM) for prediction of injury severity of construction workers. The findings of their research explored that severity level could be predicted by the elements of the working environment, such as equipment, tools, activities, physical attributes, and more importantly, environmental conditions of the workplace. Another important study carried out by Tixier et al. (2016a) in construction industry explored that prediction of injury severity is random since the two models RF and SGTB performed poor in prediction. Later in 2017, Hallowell et al. (2017) endeavoured to predict the injury severity based on energy concept and concluded that energy is an important predictor. In summary, though the major accident research focuses on the injury causation, prediction of injury severity is also of utmost importance in accident prevention. Some of the selected articles focusing on injury severity prediction are listed in Table 9.

(e) Others: Besides the main four research domains identified, there exist a few important studies related to the application of decision support system (DSS), internet of things (IoT), big data, and image/video processing used in occupational accident domain. They are briefly discussed below.

(i) Decision support system

There are a number of applications of DSS in industrial, business, and research sectors. For examples, Imriyas et al. (2007) developed a DSS for predicting occupational risk in building projects. Their developed system can monitor human safety in the construction area, and helps in deriving accident indices. Cooke et al. (2008) developed a web-based decision support tool for risk management in the construction project. Another interesting work done by Cebi et al. (2017) is the development of a web-based DSS to predict the risks at shipyards. The purpose of this development was to help the decision-makers take efficient decisions based on the safety rules obtained by this system for

Table 10

Summary of some selected articles focusing on other domains.

Domains	Author (year)	Objective
DSS-based safety analysis	Uricchio et al. (2004) Gross et al. (2013) Cebi et al. (2017) Caballero-Ruiz et al. (2017)	To evaluate the groundwater pollution risk A develop a clinical decision system to suggest appropriate rehabilitation for injured workers To develop a web-based DSS (WDSS) to predict potential risks at shipyards To develop a WDSS suggesting automatic diet prescription for the patient
Big data or IoT- based safety analysis	Wu et al. (2019)	To monitor and record working conditions of production systems and location information of underground equipment and miners
Video/Image processing- based safety analysis	Mandal et al. (2016) Yu et al. (2019)	To detect the bus driver's fatigue using visual analysis of eye state To monitor the whole-body physical fatigue of construction workers using a novel non-intrusive method with computer vision

each machine and tool used. Cheng et al. (2002) developed a DSS based on Geographic Information System (GIS) to help engineer in construction sites in controlling the condition during excavation. Their developed systems eventually improved the instrumentation program and helped to predict adverse conditions in construction. The system enabled the decision-makers to take adequate actions based on the analysis made by the system. The clinical field is not left untouched by the revolutionary impact of support systems. A clinical decision system is used to suggest appropriate rehabilitation for injured workers Gross et al. (2013) and a WDSS suggesting automatic diet prescription for the patient has been built (Caballero-Ruiz et al., 2017). In addition, few more studies in this domain have proved its utility in its practical implication (Ayed et al., 2010; Chae et al., 2003; Barjis et al., 2013). It has also a large impact on the environment. One of such studies includes a knowledge-based decision support system, which was developed for risk assessment of groundwater pollution (Uricchio et al., 2004). Therefore, it is observed that the DSS has importance in occupational accident analysis domain.

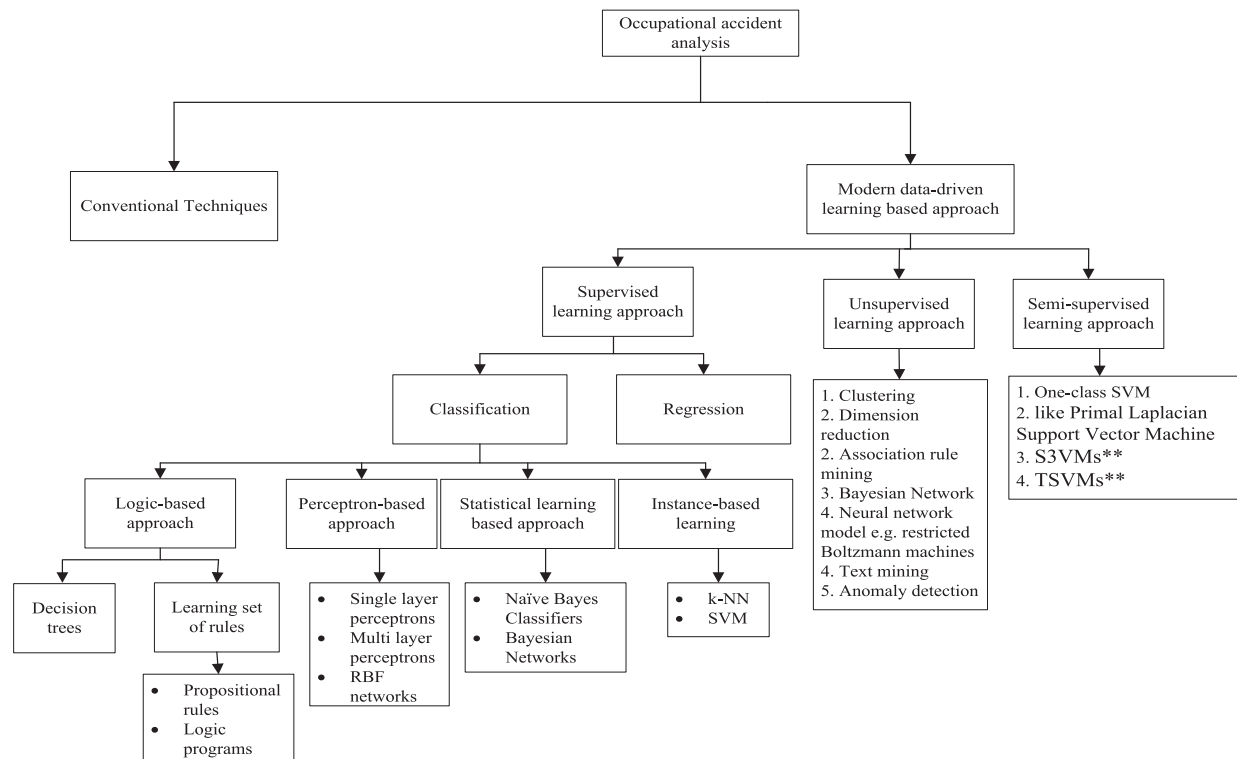
(ii) Internet of Things

Internet of Things is a computing concept that helps in communication by using wired and wireless networks, technologies, and the

elements of wireless sensors networks. Its application in the occupational incident analysis is very scarce. Based on our review, one paper published recently has been retrieved. Wu et al. (2019) established a dynamic information platform for underground coal miners. It enables easy monitoring and recording the working condition data of the production systems and the location of miners and equipment. The big data collected by this system is analysed using cloud computing techniques and safety-critical situations are identified. The developed platform offers a 3-dimensional (3D) virtual system of a mining environment, safety inspection and diagnosis system, and particularly, a safety emergency system. The overall system can spot hazardous areas and help prompt decision making for mining safety.

(iii) Big data

The term “big data” implies a collection of complex and large data set that is hard to process using on-hand database management tools or traditional data processing applications. However, the term refers not only to the huge volume of data sets, but also to the whole tools, techniques, and frameworks used to analyse it. Primarily, the big data application has been characterized by five ‘V’s, i.e., ‘Variety’, ‘Volume’, ‘Veracity’, ‘Velocity’, and ‘Value’ (Hu et al., 2014). In occupational accident research with real-time data storage facility with the massive



Note: ** TSVM: Transductive SVMs; S3VMs – Semi-supervised Support Vector Machines.

Fig. 18. The taxonomy of ML approaches.

support of cutting-edge technologies and state-of-the-art tools and techniques, it becomes possible to analyse any hazardous scenario in more detailed ways. As for an instance, real-time traffic monitoring systems as well as traffic safety, have been a research issue for many researchers (Shi and Abdel-Aty, 2015). Though there exist sufficient amount of works related to process innovation and technology advancement towards the improvement of the safety management system as well as labor productivity, especially in construction domain (Alikhani et al., 2013), warning system remains problematic (Wang and Razavi, 2015). However, the use of big data and its associated technologies in predictive modeling opens up a few real challenges along with several advantages. Using this, experts could quickly analyse a very large amount of data either in a static or real-time way. Thus, the analysis with more data with advanced techniques is the main source of accurate and precise prediction of occupational accidents. On the contrary, some issues like the privacy of big data usage, scarcity of advanced tools have plagued its superiority over other techniques.

(iv) Image and video processing

Along with the emergence of DSS, IoT, and big data, the use of image and video processing techniques in the occupational incident analysis is increasing gradually. Mandal et al. (2016) proposed a vision-based detection system for the driver's fatigue. The system has different modules, including face and eye detection, head-shoulder detection. Experimental results explored that the developed system is effective while a camera is located at an oblique position to the driver's face. Another notable work done by Yu et al. (2019) showed the development of a 3D motion capture algorithm to monitor the fatigue of workers in a construction sector. Some of the selected articles focusing on other domains are listed in Table 10.

3.4.2. Growth of research in ML approaches

Investigating the growth of research in ML approaches in occupational accident domains, there are three broad categories observed; (i) supervised learning approach, (ii) unsupervised learning approach, and (iii) semi-supervised learning approach. A taxonomy of the application of ML approaches in occupational accident analysis is shown in Fig. 18. It shows a classification of analytical methods used in this domain. There are broadly two techniques available, one is conventional techniques and other is ML-based techniques. Our research focuses on the latter. These approaches using ML approaches in this research domain are briefly described below.

(a) *Supervised learning approach*: In the supervised learning approach, there are mainly two types of analyses observed, (i) classification-based, and (ii) regression-based analysis. In the classification-based analysis, there are broadly four approaches present, (i) logic-based approach, (ii) perceptron-based approach, (iii) statistical learning-based approach, and (iv) instance-based approach. On the other hand, there are three types of techniques available in the literature under unsupervised learning approach: (i) clustering technique, (ii) association rule mining technique, and (iii) text mining or natural language processing (NLP) technique. All the above-mentioned techniques are briefly discussed below.

1. Classification

(a) Logic-based approach:

In the logic-based analysis, decision tree and rule-based learning (RBL) are available, which are discussed below.

- *Decision Tree*- Decision tree (DT) learning is an important technique in the prediction of occupational incidents. The DT learning uses DT as a predictive model which maps the observations of an item to its target value. The DT with a finite set of values for a target variable is called classification tree, and with continuous values (usually real numbers) is called regression trees. Many studies on the prediction of occupational accidents are done in different application domains like mining (Groves et al., 2007; Komljenovic et al., 2008),

construction (Amiri et al., 2016; Amiri et al., 2014; Mistikoglu et al., 2015; Rivas et al., 2011), manufacturing (Sanmiquel et al., 2015), aviation (Christopher and alias Balamurugan (2014), railways (De Ona et al., 2014; Mpinda et al., 2016) and chemical industry (Cheng et al., 2013). Many researchers have found some advantages by using DT in their respective problem domain. They are easily interpretable by humans and non-linearity can be explained by DT. Training time is also less than other classifiers though it depends on the size of data set. However, some drawbacks also are available in their use. They have high variance, and more prone to overfitting problem. Moreover, small perturbation could also lead to a major change in the DT structure.

- *Learning a set of rules/rule-based learning (RBL)*- In this method of prediction, one identifies the relations and links among input and output parameters and generates some definite rules to classify a new input. To make a set of rules one first learns a DT and then converts it into rules, or uses other algorithms to do so. The main aim is to generate the smallest set of rules consistent with training data. The popular algorithms used include PART (Frank and Witten, 1998), and sequential covering algorithms. Sequential covering algorithm was used for assessing occupational risk in the ship-building industry (Fragiadakis et al., 2014). In particular, while operating with binary problems, rule learning appears to be more comprehensible than DT as rule-based learners learn a set of rules for only positive class. But, for multi-class, the use of it adds complexity and demands more time as this approach generates a set of rules for each class separately. The interpretability of the classifiers is high. The higher classification accuracy and lower training time are realized while numerical variables are properly discretized before induction.

• Perceptron-based approach:

The term 'perceptron' means an artificial neural network (ANN) of just two layers of nodes, i.e., input layer nodes and output layer nodes. Usually, perceptron-based approaches can be classified as single-layered, and multi-layered. There are many single-layered perceptron-based algorithms like WINNOWER (Littlestone and Warmuth, 1994), voted perceptron (Freund and Schapire, 1999). The advantages of single-layered perceptron include superior time complexity while dealing with irrelevant features. A multi-layered neural network consists of input, output and hidden layers. The working principle is as same as single the layered perceptron except it has a number of hidden layers. However, the accuracy of MLP can be seriously affected by the optimal choice of the number of hidden layers and the number of hidden nodes per hidden layer. Another class of ANN is a radial basis function (RBF) that is characterized by three-layered feedback networks (Durduran, 2010). The drawbacks of RBF include the difficulty in the selection of an optimal number of basis functions which controls the complexity and generalization ability of the network. In accident research, there is substantial evidence of the use of ANN by many researchers. van den Honert and Vlok (2015) used ANN to identify patterns between input attributes and to predict the risk of accidents occurring. A predictive ANN model was created, trained, and validated with the help of a case study in a platinum mine near Rustenburg in South Africa. Saini and Yousif (2012) implemented the Multilayer Perceptron (MLP) neural network topology in mishap analysis and prediction. Fragiadakis et al. (2014) used an ANFIS model based on both ANN and fuzzy logic (FL) to predict occupational accident risk in the shipbuilding industry.

(b) Statistical learning-based approach:

In this approach, two types of methods are used, (i) Bayesian

network, and (ii) Naive Bayes. They are briefly described below.

- **Bayesian networks (BN)**- Bayesian network (BN) is the most widely used technique in a statistical learning paradigm. A BN is a graphical model that can be represented as a directed acyclic graph (DAG) with a finite number of nodes. It can show the probability relationships among variables (features) connected with arcs representing causal influence among features. The main task of learning a BN is twofold; first to learn a DAG structure, and second is to estimate the parameters. Many researchers are found to be using BN for various problem solving, including predictive modeling. Many applications are found in different sectors where BN was used as a prediction model of accident occurrences. [Ozbay and Noyan \(2006\)](#) discussed the estimation of incident clearance times using BN. They analysed stochastic variations of the data and bi-directional induction in decision making through modeling by BN using incident data from Northern Virginia. In one hand, BN serves as powerful modeling techniques to analyse the cause-effect relationship of any incident ([Baksh et al., 2015](#)) and on the other hand, it can perform satisfactorily as a predictive modeling tool for accidents. [Martin et al. \(2009\)](#) showed that the BN performs well in cause-effect analysis and prediction of incidents related to falls from ladders in workplaces. Though the results of dynamic BN is found to be compromised with noisy environments to some extent ([Zhao et al., 2015](#)), it can be used efficiently to diagnose the faults and to predict the accident causes. Many recent studies use BN to figure out the most probable causes of operational risk ([Bolsover, 2015](#); [Vieira et al., 2015](#)). In the context of risk management in the industrial domain, BN has been used for better decision making through effective predictive modeling of reliability, availability, maintainability, and safety, i.e., RAMS ([Zhang et al., 2014](#)). Another important application is to incorporate the fuzzy concept in different occupational accident domain like near-miss ([Taylor et al., 2014](#); [Wellman et al., 2004](#)).
- **Naïve Bayes**- Naïve Bayesian (NB) network is the simplest form of BN comprising one parent node and several child nodes in the structure with a strong assumption of independence among the child nodes. Many researchers use this algorithm along with other algorithms. In the occupational safety analysis, very a smaller number of studies has been conducted using Naïve Bayes (NB) classifiers. [Chen et al. \(2016a\)](#) discussed an explanatory analysis of driver injury severity in rear-end crashes using a decision table/Naïve Bayes (DTNB) hybrid classifier. Another important application of NB classifier is to replace human efforts for coding a large amount of data set which is regarded as a resource-intensive and unavoidable task ([Bertke et al., 2016](#); [Nanda et al., 2016](#)). NB performs well with increased accuracy while operated on balanced data sets created by oversampling techniques ([Mujalli et al., 2016](#)).

(c) **Instance-based learning approach:**

Instance-based learning is a set of learning algorithms. It compares new instances with the instances observed in training, which are stored in memory. That's why it is also called a memory-based learning approach. *k*-Nearest Neighbour (*k*-NN) ([Goh and Ubeynarayana, 2017](#)), the kernel-based algorithms, such as SVM, RBF networks are the main instances of this kind of learning approach. It is a lazy learning approach as the generalization is delayed until a query is made to the model or system. It usually constructs hypotheses from training instances. Consequently, it augments the complexity of the hypotheses in the order of $O(n)$ where n is the number of training items within a hypothesis. In literature, instance-based learning approach uses SVM mostly. Using SVM, many studies on occupational

accident prediction are carried out by many researchers, including [Sánchez et al. \(2011\)](#), [Matias et al. \(2008\)](#). The main advantage of the using instance-based learning approach is that it has the capability of adapting its model to historical unknown data.

2. **Regression**

A regression model is used to establish a relationship of one or more dependent attributes with a set of explanatory or predictor variables, also known as independent variables (IVs). In most of the cases, it is found in the literature that the DVs are usually categorical in nature; researchers often prefer logistic regression as the relationship between the accident categories and one or more covariates or predictor variables could be quantified ([Ciarapica and Giacchetta, 2009](#)). Apart from these two, other linear models are the analysis of variance (ANOVA), analysis of covariance (ANCOVA), probit, log-linear, Poisson, and negative binomial regression. All these models are from a family of linear models called generalized linear models (GLM). The detailed description of GLM can be found in [McCullagh and Nelder \(1989\)](#). The application of GLM, particularly logistic regression was extensively used in mine accident modeling ([Maiti and Bhattacharjee, 1999](#); [Onder and Mutlu, 2016](#); [Palei and Das, 2009](#)).

(b) **Unsupervised learning approach:** In unsupervised learning approaches, there are mainly three techniques used, (i) clustering, (ii) association rule mining, and (iii) text mining or text analysis, which are discussed below.

1. **Clustering techniques**- Clustering is an unsupervised learning approach with the main function is to club or group the data objects into a finite number of different clusters having strong similarity among the members within a group. Though the main function of clustering is not a prediction, it can be used in building predictive modeling. Clustering of data from higher dimensional spaces is an important task since they discover the latent structures in data. Among other clustering techniques, *k*-means clustering is reported to be widely implemented in different incident areas ([Hajakbari and Minaei-Bidgoli, 2014](#); [Palamara et al., 2011](#); [Polat and Durduran, 2012](#)).
2. **Association rule mining techniques**- Like clustering algorithm, association rule mining (ARM) is an unsupervised learning algorithm with the main purpose to find the associations or interesting relations among the input variables towards any definite outcomes. It is intended to figure out the strong rules among many others generated in databases. But, its functionality can be extended towards building a classification task or predictive modeling ([Hu et al., 2002](#); [Wang and Wong, 2003](#)), and thus can be used in different incident areas, including construction ([Cheng et al., 2010](#)), traffic ([Soysal, 2015](#); [Wen et al., 2019](#)), and steel industries ([Verma et al., 2014](#)).
3. **Text mining/text analysis**- The narrative text is one of the key resources for the prediction of an accident. It provides valuable additional information in the analysis along with other types of data. A few studies have been conducted for accident analysis, which justified the use of text data in different applications areas, for example, construction industry ([Tixier et al., 2016a](#); [Tixier et al., 2016b](#); [Tixier et al., 2017](#)), fire services ([Taylor et al., 2014](#)), oil industry ([Sanchez-Pi et al., 2016](#)), aviation industry ([Andrzejczak et al., 2014](#); [Koteeswaran et al., 2017](#); [Kuhn, 2018](#); [Tanguy et al., 2016](#)), road ([D'Andrea et al., 2015](#); [Pereira et al., 2013](#)), rail ([Brown, 2015](#); [Li et al., 2018](#)), and so forth. [McKenzie et al. \(2010\)](#), [Vallmuur et al. \(2016\)](#) showed the significance of text analysis in accident research through an extensive review of the literature. To investigate the importance of narratives in the prediction of occupational accidents, [Li and Guo \(2015\)](#) tried to analyse aviation safety data with the help of topic modeling techniques. Related to this, a noteworthy contribution made by [Brown \(2016\)](#) is to analyse

rail accident data to explore the main contributors behind the accident using text mining associated with other techniques, such as Latent Dirichlet Allocation (LDA), and Random Forest. Thus, the main challenge lies in the analysis of the unstructured text. To tackle the issue, Tixier et al. (2016a) tried to develop a system that could overcome the problem by decoding the unstructured text from the accident database. The system developed by them could use the unstructured injury database with 101 attributes, and produced with 95% classification accuracy. Vallmuur et al. (2016), therefore, mentioned in his study that future research on injury analysis would direct a continued growth and advancement in the application of text mining for utilizing information within the texts. The primary difficulty in the analysis of the free unstructured texts is the sparsity and high-dimensionality of a document-term matrix. Moreover, a text mining-based approach cannot capture the order of words and semantic meaning of them. One recent study by Pavlinek and Podgorelec (2017) shows that topic modeling of free-text could help in text classification task and reduction of sparsity. In road accident analysis, one study by Pereira et al. (2013) used topic modeling of incident reports of traffic to extract information in real time to predict the incident duration.

(c) *Semi-supervised learning*: There are scanty works available on the prediction of an occupational accident by semi-supervised learning approach. From the literature, some semi-supervised techniques like Primal Laplacian SVM, one-class SVM have been used in different domains of occupational accident research like near-miss detection (Yang et al., 2016).

3.5. Summary

From this comprehensive four-stage review process, some key findings from the analyses on 232 articles are obtained, which are given below.

- (a) From *descriptive analysis*: (i) an increasing trend of ML approaches is observed from 1995 to 2019 (up to May). However, a steep fall in trend is observed in 2014; (ii) Four journals, namely 'Safety Science', 'Accident Analysis & Prevention', 'Automation in Construction', and 'Journal of Safety Research' are found to be the topmost journal in this domain, which have contributed 18.97%, 15.51%, 6.47%, and 4.74%, respectively; (iii) non-tree-based algorithms (44.83%) are used more frequently as compared to tree-based algorithms (27.16%), text mining-based algorithms (11.64%), and decision support system (3.02%); (iv) the authors, Maiti J., Abdel-aty M., Li Heng are ranked top in the authors' list publishing research articles in ten, five, and four in numbers, respectively; (v) three institutes, namely IIT Kharagpur, HKPU, and National University of Singapore have contributed significantly with 3.17%, 2.46%, and 2.46%, respectively; (vi) three countries/regions, namely USA (74), China (38), Taiwan (15) are at the forefront in publishing research articles; (vii) among the sponsoring agencies, 'National Natural Science Foundation' in China is ranked in the top of the list; (viii) research has been more carried out on road accident with 36.64% contribution, which is followed by construction accidents (21.98%), mining accidents (6.90%), and others.
- (b) From *scientometric analysis*, (i) the journal 'Soft Computing' is identified as the most influential journal in this domain; (ii) three journals, namely 'Soft Computing', 'Computers and Industrial engineering', and 'Injury control and safety promotion' have been publishing articles actively; whereas 'Ergonomics', and 'Risk Analysis' are not regularly publishing articles in this domain; (iii) Some keywords, including 'Text mining', 'Ontology', and 'Decision tree' are found to have more impact in this domain and used more frequently; (iv) the countries like USA, China, Australia, Germany,

Canada are strongly connected, showing collaborative research work among them.

- (c) From *citation network analysis*, a total of five clusters (i.e., domain-clusters) are identified. They are: (i) prediction of incident outcomes, (ii) rule-based pattern extraction and prediction of incidents, (iii) risk prediction, (iv) injury severity prediction, and (v) others, which focus on the application of DSS, image or video processing or IoT-based incident analysis.
- (d) A taxonomy is obtained, which explores the classification of ML approaches used in the occupational accident analysis.

4. Research issues and scopes for future works

Reviewing the previous studies through a four-phase review process, several research issues and scopes for future works are identified. Research issues are classified according to (a) data pre-processing-related, (b) data source-related, and (c) algorithm-related issues, which are elucidated as follows:

(a) *Data pre-processing-related issues*: In data pre-processing related issues, there are mainly four types of issues observed in (i) unstructured text data handling, (ii) missing data handling, (iii) data uncertainty handling, and (iv) class imbalance handling, which are briefly discussed as follows:

1. **Unstructured text data handling**: The performance of the classifiers not only depends on optimized parameter values but also the types of data used. As it is a known fact that numerical attributes hold more information than categorical attributes or free-text attributes, thus dealing with different data types also impacts classifier's performance. Hence, it is really a challenging task for researchers to extract a pattern from different types of data like categorical or more specifically, unstructured free-text data. Most of the literature in accident domain used either numerical data or categorical data for accident analysis; whereas, analysis of free-text data often remains unutilized or under-utilized as it is eventually a difficult task to extract the pattern from the unstructured passage of texts. Therefore, a focus should be given to utilize the hidden information within texts in accident analysis and prevention.
2. **Missing data handling**: Missing values in data are common in practice. This happens due to erroneous data logging into a system or the system error, or human error. Data analysis without handling this issue often leads to erroneous or sub-standard or misleading results. There are a number of methods used for predicting or imputing missing entries in the dataset, for example, hot-deck imputation, mean imputation, k-NN imputation, expectation-maximization (EM), regression-based imputation, random forest (RF)-based imputation, and so forth. (Stekhoven and Bühlmann, 2011; Baraldi and Enders, 2010; Hernández-Pereira et al., 2015; Xia et al., 2017). Although this pervasive issue is reported in different domains, including road safety (Imprialou and Quddus, 2019), and healthcare (Wolfson et al., 2015; Hernández-Pereira et al., 2015), it has rarely been addressed in the occupational accident analysis. Therefore, an in-depth investigation of missing data handling in this domain is further required.
3. **Class-imbalance handling**: Besides the issues of unstructured text data and missing data in accident analysis, there persists another challenge, called *class imbalance*. It occurs when the number of one category of the response attribute is much higher than that of other categories, which results in a high skewness in class distribution. The prediction under such circumstances becomes biased towards the majority class and may yield higher accuracy, while it may not predict the single minority class accurately. Besides, due to the presence of class imbalance, "*small disjuncts problem*" arises, which is occurred during the erroneous rule induction due to the presence of fewer instances (Douzas et al., 2018). To address this issue, there exist many techniques, which are mainly classified into three broad

categories: (i) data-based, (ii) algorithm-based, and (iii) cost-sensitive-based approaches. Among them, data-based approaches are capable of generating more balanced dataset (Douzas et al., 2018). These approaches include undersampling and oversampling algorithms. Unlike the former algorithms, the latter can overcome the risk of information loss in data efficiently and attains more popularity in overcoming the issue of data imbalance. A series of improvement in oversampling techniques have emerged. Some of the popular and powerful techniques among them include Synthetic Minority Over-sampling Technique (SMOTE) (Chawla et al., 2002), Borderline SMOTE (BLSMOTE) (Han et al., 2005), Majority Weighted Minority Oversampling Technique (MWMOTE) (Barua et al., 2012), *k*-means SMOTE (KMSMOTE) (Ma and Fan, 2017). However, the number of studies which addressed the data imbalance issue in accident analysis are reportedly very less (Jeong et al., 2018; Poh et al., 2018; Zhu et al., 2018). Therefore, an in-depth investigation of class-imbalance handling in this domain could be another interesting topic of research.

4. **Data uncertainty handling:** Another difficulty in data include uncertainties, which are often occurred due to human errors, or by means of process complexity. If there exists a dataset having a substantial amount of these issues, the desired results cannot be achieved. Therefore, a proper data pre-processing technique is required, which can overcome this shortcoming. Rough set theory (RST) is one of the popular data reduction techniques and is very useful in the handling of data uncertainty. Because of the advantages of RST, it has been successfully used in different application areas, including railway (Jiao and Sun, 2008), and construction (Amiri et al., 2017). However, its use is rather limited in our research domain to handle the above-mentioned issue. Therefore, a new methodology needs to be developed for uncertainty handling in data.

(b) *Data source-related issues:* In accident analysis, reactive data are mostly used (Chen and Chen, 2011; Chen and Luo, 2016; Delen et al., 2017; Hallowell et al., 2017; Poh et al., 2018; Tixier et al., 2016a). By definition, reactive data indicates the data collected after the occurrence of an incident, for example, injury frequency, lost workdays, compensation costs, etc. On the contrary, there exists another kind of data, which is captured before any incident takes place. It is called proactive data, for instances, inspection report, safety audit report, safety training report, etc.

Although the use of reactive data has gained increasing attention in accident analyses, it faces a serious drawback. During the analyses of reactive historical data, the development of the immediate injury is initiated already. The lag in analysis eventually marks a blind spot which decision-makers hardly have any perception regarding the factors leading to the occurrence of injury. The analysis of proactive data helps the decision-makers like safety professionals take proactive measures by observing the progression of factors leading to accidents occurring on a real-time basis. In fact, the analysis of such data can effectively provide a warning sign prior to the occurrence of injury, which often remains unnoticed. At the earlier stage of such occurrences, signs are perceived as noise. Hence, it is pertinent for the analysts to identify the signs from the existence of noise that could help to predict the occurrence of accidents. Although there persists a common notion that leading indicator-based proactive data analysis can be effective for prediction, a substantial effort has not been yet put forward (Khawaji, 2012) due to unavailability of effective leading indicators across industries. In addition, according to Rasmussen (1997), major accidents leading to injury take place, not because of just the existence of physical events, but the relaxation of safeguards and controls. If the belief is deemed to be true, there must be a pattern to be identified, which leads to an occurrence of accidents. Hence, it is essential to utilize proactive data in accident prediction.

In support of the use of proactive data, some previous studies have

mentioned their utilities. To exemplify, Grabowski et al. (2007) reported that proactive data are useful to analyse the various factors like conditions or events that result in an injury and hold predictive value. According to Hinze et al. (2013), proactive data act as the building block of the safety culture of any organization. On the other hand, there is an increasing number of queries related to the value of predictive outcomes of reactive data and their usability in mitigating injuries. Therefore, safety cannot be ensured by depending on reactive data, rather it needs a constant focus on the combination of both reactive and proactive data (Reiman and Pietikäinen, 2012). Indeed, depending on an individual kind of data may be less significant while it is used in isolation. However, it turns out to be important while used in combination. Realizing the significance of the nature of data, Reiman and Pietikäinen (2012) remarked that the utilization of both reactive and proactive data in analyses is eventually useful for any organization. Some recent studies by Fite et al. (2010), Sarkar et al. (2020) have shown the utilities of using both types of data in the analysis. Therefore, in incident analysis, the use of both data is effective and should be used simultaneously.

(c) *Algorithm-related issues:* In algorithm-related issues, some challenges pertaining to parameter optimization of classifiers and crisp rule extraction are briefly discussed below.

1. **Parameter optimization of classifiers:** All the machine learning algorithms do not provide optimal results, such as classification accuracy and understandability/interpretability, if the parameters of them are not properly tuned or optimized. To optimize their parameters, optimization methods are found to be most useful than other techniques like manual tuning or grid search. In occupational accident research, hardly any study or no study has been reported that uses optimization techniques on classifiers to obtain better classification accuracy. Apart from the traditional gradient descent (GD) method, there are a number of popular meta-heuristic optimization techniques available, including particle swarm optimization (PSO), GA, ant colony optimization (ACO), and artificial bee colony (ABC), which are used for parameter optimization (Nieto et al., 2015; Rashno et al., 2017). The main reason for using such algorithms is to obtain a global optimum solution regarding fitness function. Of them, PSO, and GA have gained widespread popularity and have been used in a broad spectrum of application domains (Tsoukalas and Fragiadakis, 2016; Chau, 2007; Sarkar et al., 2019b). For example, it is observed that to get enhanced accuracy in the SVM model, penalty factor (*c*), and kernel parameter are considered to be optimized (Chou et al., 2014). There are many optimization techniques used for this purpose like a GA, PSO, gradient descent method, etc. (Nguyen et al., 2015). For ANN, there exist several parameters which can be optimized, such as the number of layers, input and hidden neurons, type of transfer functions, the topology of ANN, weights, threshold, and others. Li and Guo (2015) used initial parameters, network topology, weights, and thresholds of BPNN based on a memetic algorithm with GA. Xue and Liu (2017) used only initial weights and threshold values for the back-propagation model for predicting liquefaction susceptibility of soil. Das et al. (2014) focused on optimizing the weights, transfer functions, and topology of ANN for channel equalization. Among the decision tree algorithms, the two parameters of RF, 'ntree' (the number of trees in the RF), and 'mtry' (the number of predictors considered at random at each split) are optimized by grid search algorithm (Tixier et al., 2016a; Tixier et al., 2016b). Some studies used parameter optimization of CART, C5.0 algorithms also. The parameters of CART, namely 'complexity parameter (cp)' (Han et al., 2007), 'minimum number of observations that must exist in a node in order to enable split (minsplit)' (Mantovani et al., 2018; Shine et al., 2018), and 'the maximum depth of any node of the final tree (maxdepth)' (Mantovani et al., 2018; Shine et al., 2018; Sarkar et al., 2019a) are optimized. On the other hand, the parameters of C5.0, namely

'minimum cases for the best solution (mincases)' and 'confidence factor (cf)' are optimized in the study of Munoz-Mas et al. (2016). For the ANFIS algorithm, premise and consequent parameters are optimized using a gradient descent algorithm (Fragiadakis et al., 2014). All of the above-mentioned studies showing the importance of optimization techniques on classifiers have been carried out to solve the problems in other domains. Nevertheless, its use in occupational accident analysis domain is still new. Therefore, parametric optimization of the classifiers for better prediction performance is required to be further studied.

2. **Crisp rule extraction:** Although the development of the model for data reduction and prediction is important, it may not be sufficient for the management perspective. A list of implementable recommendations is, in this case, effective for the management. Extraction of rules from data is, therefore, an efficient way of providing such recommendations. However, the rules generated from data often suffer from uncertainties or impreciseness, which result in misleading or erroneous decision making. To overcome this issue, the rough set approach (RSA) can be a better choice of the researchers. This approach is capable of extracting crisp rules from the data in different application domains, for example, construction (Tam et al., 2006), road (Wong and Chung, 2008), and so forth. However, RSA based on traditional indiscernibility relation is unable to deal with the issue of incompleteness in data. Therefore, crisp rule generation and interpretation using the above-mentioned approaches are required to be further investigated.

From the overall study presented in this paper, several scopes for future research works can be determined. For the ease of understanding, they are categorized according to (i) implementation, (ii) empirical observation, and (iii) theoretical developments points of view, which are summarized below.

From the *implementation point of view*, the following works can be done:

- (i) An advanced decision support system can be developed for real-time data analysis and safety-related decision making.
- (ii) A cloud-based model can be developed and embedded into the system for the storage of big data, which aids in efficient data analysis, and quick decision-making.
- (iii) A video surveillance system can also be implemented in the workplace to monitor the safety violation or anomaly detection and accordingly, proper actions can be taken to mitigate the occurrence of incidents.
- (iv) Internet of Things-based safety analysis can be another promising domain of research.

From the *empirical point of view*, the following research inquiries can be undertaken:

- (i) Rule-based pattern extraction from data for other incidents can be done.
- (ii) Leading and lagging indicators can be developed for a safety study.
- (iii) A number of different reactive and proactive data can be collected and used together for the prediction and analysis of incidents.
- (iv) To obtain better generalizability of results, a large number of safety-related data should be collected within and across different industries.

From the *theoretical development point of view*, the following works may be undertaken:

- (i) This research has used the unstructured incident text or narratives for incident analysis. A critical area of future research could be the identification of reliable words and the inter-relationships among them to capture the sentiment of the safety professionals who log

the data into the system.

- (ii) For minimizing effort and time for text pre-processing and categorization, an ontological concept can be used.
- (iii) From short incident texts, finding a latent topic can be another future research area.
- (iv) The class imbalance is a big issue in data analysis. Various popular oversampling algorithms are used to alleviate the issue for making the prediction unbiased. As future studies, a comparative analysis using various oversampling techniques, undersampling techniques, algorithm level techniques, and cost-sensitive techniques can be carried out.
- (v) Development of dynamic predictive and prescriptive modeling for a data stream or real-time data analysis is another important future research direction.
- (vi) For effectively handling fuzziness in data, dominance-based rough fuzzy sets can be incorporated. This can be useful for both data pre-processing stage and the extraction of crisp rules from data.

5. Conclusions

This study uses a four-phase review of state-of-the-art on the application of machine learning (ML) techniques in occupational accident analysis. The review process involves four phases of analyses, namely bibliometric search, descriptive analysis, scientometric analysis, and citation network analysis (CNA). In the bibliometric search, a total of 232 articles are systematically extracted from 1995 to 2019 (up to May). Based on these articles, descriptive analyses are done to identify the distribution of the number of articles by the year of publication, by different journals, by data collection method, by different methods, by authors' contribution, by the contribution of different institutes, by different countries/regions, by different sponsoring agencies, and by different industries/sectors. After the descriptive analysis, a scientometric analysis is conducted to find the influences of journals, authors, authors' keywords, articles/documents, and countries/regions. Thereafter, the CNA is carried out to classify the overall research domain according to the research themes and methods used. From this extensive review, some of the key findings are summarised. First, from the analyses, it has been observed an increasing trend in the application of ML approaches in occupational accident analysis during the above-mentioned period. Second, the journal 'Safety Science' is found to be the leading journal in this domain, which has been publishing research articles in this domain more frequently. Third, it has been observed that three countries/regions, namely USA, China, and Taiwan are leading in publication. Fourth, a total of five research domain clusters are identified using CNA, and finally, a taxonomy of ML algorithms used in this domain is obtained. Finally, several important research gaps or issues identified as the outcomes of this literature review and the scopes for future work are discussed.

Like other research, the present research has some limitations. For examples, the study considers the span of years 1995 to 2019 (up to May). There could be other related studies beyond this range. In addition, the articles used in this analysis are retrieved based on the selected keywords used. If there were other related keywords, there would have been other related articles obtained. Moreover, the study considers only the articles published in English. Thus, the study might have excluded potential and relevant papers published in other languages. Besides this, the study also ignores the important information of the studies published as either conferences or books or book chapters (if any).

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